

Highlights

The continuum limit of a space-filling layer of cubes: a tiling identity, an exact lattice kernel and the closed-form error of the discrete layer

T. M. Nestor

- Voxel coupling averaged over the source cell equals the continuum's exactly
- The cube's single-site T-matrix follows from distributional moment integrals
- The voxel scheme's leading error is known in closed form for every grid
- A first-moment voxel converges at fourth order for P, SV and SH waves
- In a random stratified layer, one voxel per model layer reaches 1e-6 accuracy

The continuum limit of a space-filling layer of cubes: a tiling identity, an exact lattice kernel and the closed-form error of the discrete layer

T. M. Nestor

Independent researcher, Australia

Abstract

We analyse the convergence of voxel discretisations of the elastic volume integral equation, in which a medium is divided into cubic cells coupled through the background Green's tensor: the elastic coupled-dipole method. Because cubes tile space, n planes of voxels of side D/n are the layer itself for every n , with no shape error, so the difference from the exact solution is the discretisation error alone. We show that (i) the cube's lateral form factor vanishes on the reciprocal lattice, so the source-cell-averaged coupling equals the continuum's; (ii) the single-site T -matrix follows from distributional moments of the Green's tensor and, against exact scattering by a sphere, is exact statically and departs only by a real, closed-form $(ka)^2$ term; (iii) its self term cancels, so the scheme is a collocation of the continuum equation, of second order with leading error $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission, which no scalar correction can remove; (iv) a Galerkin voxel carrying Legendre moments of its internal field to degree p converges at order $2p + 2$, with leading error $-(kd)^4/720$ at $p = 1$, for incident P, SV and SH waves at normal and oblique incidence, and one plane of it reproduces the

thin-layer series through $O(D^{2p+2})$. In a randomly stratified layer the orders are unchanged, and the fourth-order voxel reaches an accuracy of 10^{-6} with one voxel per model layer. A smoothly graded sphere converges at second order against its exact solution. Every result is verified by an independent implementation or an exact solution.

Keywords: Elastic wave scattering, Volume integral equation, Foldy–Lax multiple scattering, Discrete dipole approximation, Galerkin method, Convergence analysis

1. Introduction

The problem this paper serves. Seismic waves in the crust are scattered by heterogeneity on every scale. The models that describe it are parameterised in many ways (as layers or blocks, by smooth basis functions, or by interfaces), and a calculation that uses one must first sample it onto a grid. A volume-integral scattering calculation represents the sampled model as cells of constant properties and treats every cell as a scatterer: a voxel carrying a single-site T -matrix and coupled to every other voxel through the Green’s tensor of the background, with the multiple-scattering series summed. Where the medium varies within a cell, that sampling is itself an approximation: §9 shows that it caps any voxel at second order, unless the cell’s contrast is expanded to the degree of its field. For the media such a scheme is built for, there is no exact solution, and that is the difficulty. Its error cannot be measured against the answer. Comparison with another numerical method mixes the errors of both. And a refinement study can mislead, because a voxel scheme can converge smoothly to the wrong limit. §4 shows how that hap-

17 pens with the most natural choice of coupling, the Green's tensor evaluated
 18 between voxel centres: it carries a local term whose size per unit volume does
 19 not shrink as the voxels do, and unless the single voxel's response is matched
 20 to cancel it, refinement converges to the wrong answer without revealing it.

21 *Why a homogeneous layer.* This paper therefore starts from the simplest
 22 configuration in which a voxel scheme runs in full (every voxel, every coupling,
 23 the whole multiple-scattering series, at normal and at oblique incidence) and
 24 the exact answer is nevertheless known: a layer filled with identical voxels.
 25 Because cubes tile space, such a layer does not approximate a homogeneous
 26 layer; it *is* one. Any difference between its reflection and transmission and
 27 those of the continuous layer is therefore error of the discretisation alone, with
 28 no modelling error and no shape error behind which to hide. A sphere, the
 29 usual test object for volume-integral and discrete-dipole methods [1], cannot
 30 give this: its error mixes the discretisation with the staircase approximation
 31 of a curved surface by cubes. The layer plays the part of the patch test in
 32 finite elements: reproducing a uniform medium correctly is necessary, though
 33 not sufficient, for a scheme to be trusted in a heterogeneous one. What the
 34 layer establishes about the scheme's two ingredients, the coupling between
 35 voxels and the response of a single voxel, are properties of the ingredients,
 36 which are the same in any medium: that the averaged coupling is exact, that
 37 the self term cancels, and the form of the single-site error. §9 confirms that
 38 they carry over to a stratified medium; a laterally heterogeneous one remains
 39 to be tested.

40 *What this paper adds.* Set against the coupled-dipole method described be-
 41 low, the contributions are these. (1) *An exact coupling.* The integrated

42 Green tensor and the filtered coupled dipoles approximate the field of a cell;
 43 for cubes the source-cell average is exact, because the cube's form factor van-
 44 ishes on the reciprocal lattice (the tiling identity, §5), and at normal incidence
 45 the lattice kernel is in closed form with no lattice sum. (2) *No polarisability*
 46 *to choose.* The self term the single site carries is the one the averaged kernel
 47 omits, so the discrete system is a collocation of the continuum equation (§6).
 48 The polarisability, a modelling choice in the coupled-dipole method, is here
 49 a derived object: the moments that derive it, taken as distributions, yield
 50 Eshelby's δ term without adding it by hand, the cube's departure from the
 51 sphere as a pure cubic anisotropy, and the closure's error against exact scat-
 52 tering by a sphere as a real $(ka)^2$ term in closed form. (3) *The error in closed*
 53 *form, for every grid:* $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission,
 54 so that a grid can be chosen for a required accuracy before the calculation
 55 is run; and the opposite signs, which explain why no correction of the single
 56 site can improve both (§11). (4) *Higher order.* A voxel carrying Legendre
 57 moments of its field to degree p converges at order $2p+2$ with a closed-form
 58 constant; its first-moment case converges at fourth order for P, SV and SH
 59 waves at normal and oblique incidence and in a randomly stratified model
 60 (§9), and for a reflection accuracy of 10^{-6} it allows voxels 58 times larger in
 61 each direction than the uniform one (§11). Where the medium varies within
 62 a cell, with the contrast expanded to degree r , the order is $\min(2p+2, 2r+2)$.
 63 (5) *The elastic state.* A voxel carries a force and a moment, so its state has
 64 nine components, the coupling is a 9×9 tensor, and its local term is the
 65 depolarisation of a plate. (6) *A sphere without a staircase,* whose contrast
 66 falls smoothly to zero at its surface, so that the scheme can be tested against

an exact three-dimensional solution: it converges at second order here (§10) and at fourth order in the companion paper [2].

Why cubic voxels. Cubes are the natural cell for three reasons. They *tile space*, so a uniform region discretised into cubes is still that uniform region, whereas spheres or point scatterers can only approximate it and carry a packing error of their own. Their *regular lattice* makes the interaction between cells translation invariant, so the coupling is a convolution (Toeplitz between planes, Bloch-periodic within them), and the whole system is solved by fast transforms. And a cube small compared with the wavelength has a *single-site response in closed form*, the elastic depolarisation of a cube (§6), so the only approximation a voxel introduces is the internal field it assumes.

The coupled-dipole method. Replacing a heterogeneous body by a lattice of small scatterers that interact through the background Green’s function is the multiple-scattering description of Foldy [3] and Lax [4]. In electromagnetics it became a numerical method as the coupled-dipole method, or discrete dipole approximation (DDA): each cell of a cubic grid is a polarisable point dipole driven by the incident field and by the fields of every other dipole [5, 6]. The method is a collocation of the volume integral equation for the electric field, with the cell’s self-interaction absorbed into its polarisability [7], and Yurkin and Hoekstra [8] review it in that form. Its accuracy turns on two ingredients, and this paper takes up both for elasticity. The first is the polarisability. The Clausius–Mossotti value is the static response of a sphere; the lattice dispersion relation of Draine and Goodman [9] adjusts it so that an infinite lattice propagates waves as the continuum does to a given order in kd ; Rahmani et al. [10] instead derive it from the exact static solution for

92 the whole body, so that the method is exact in the long-wavelength limit.
 93 The second is the coupling. Integrating the Green tensor over the source
 94 cell rather than sampling it at the centre improves the accuracy, most of
 95 all for strong contrasts [11], and the filtered coupled dipoles of Piller and
 96 Martin [12], revived by Yurkin et al. [13] for large refractive indices, replace
 97 the point dipole by a source of finite extent whose spectrum is band-limited
 98 to the grid. The coupling summed over a lattice extends the method to
 99 periodic structures [14]. Its error converges at second order in the dipole
 100 spacing when shape errors are absent [1], and for a cube, the one shape
 101 a cubic grid represents without a staircase, Yurkin and Kahnert [15] reach
 102 relative uncertainties of 10^{-7} to 10^{-3} by extrapolation in the grid. The layer
 103 studied here is the elastic counterpart of that cube: a body the grid represents
 104 exactly, so that the discretisation error can be isolated. What this paper adds
 105 is the closed form: for cells that tile space, the coupling that makes the lattice
 106 reproduce the continuum exactly, and the order and the constant of the error
 107 that remains.

108 *Its elastic counterparts.* The elastic problem has the same volume integral
 109 equation [16], and every cell-based solver of it is a coupled-dipole method
 110 whether or not it carries the name. Nestor [17] divided a two-and-a-half-
 111 dimensional medium into cells that interact through the background Green's
 112 tensor, each carrying a contrast renormalised by its own near field, with
 113 the multiple-scattering series summed. Kanaun and Levin [18] discretise the
 114 inclusion problem with Gaussian approximating functions whose matrix ele-
 115 ments are analytic. Yang et al. [19] solve a stress-velocity form of the integral
 116 equation on a grid by conjugate gradients with fast Fourier transforms, and

117 Touhei [20] does the same for a half space with a generalised Fourier trans-
 118 form. Jakobsen [21] and Jakobsen et al. [22] build the seismic forward and
 119 inverse problems on the transition operator of the discretised Lippmann-
 120 Schwinger equation, in the anisotropic elastic case with a nine-component
 121 state vector. The elastic problem adds a structure the scalar and electro-
 122 magnetic ones lack. A cubic voxel responds to the incident field with a force
 123 (through its density contrast) and a moment (through its stiffness contrast),
 124 so the state at a voxel has nine components (displacement and strain), and
 125 the single site's self-interaction is the elastic depolarisation of a cube, the
 126 discrete counterpart of the inclusion tensor of Eshelby [23].

127 *This paper.* It extends the formulation of Nestor [17] to cubic voxels. §2 sets
 128 out the discrete medium and §3 the continuous layer it must reproduce. §4
 129 shows why the point coupling fails, and §5 why the source-averaged coupling
 130 succeeds exactly. §6 derives the single-site T -matrix from moment integrals,
 131 tests it against the exact sphere, and shows that with it the discrete system is
 132 a collocation of the continuum equation. §7 gives that collocation's second-
 133 order error in closed form, and §8 the fourth-order voxel, for incident P
 134 and S at normal and oblique incidence, and its generalisation to Legendre
 135 moments of any degree. §9 repeats the tests on a randomly stratified layer
 136 and on a medium that varies within a cell, §10 on a sphere, and §11 discusses
 137 what the results imply for voxel schemes in general. Every result is derived
 138 symbolically in Mathematica and checked against an independent Python
 139 implementation or an exact solution.

140 2. Setting

141 *The discrete medium.* Space is tiled by cubes of side d (half-side $h = d/2$).
 142 Cube i carries a contrast $\Delta\lambda, \Delta\mu, \Delta\rho$ from an isotropic background with
 143 Lamé parameters λ, μ , density ρ , P and S speeds α, β and P-wave modulus
 144 $M_P = \lambda + 2\mu$. Time dependence is $e^{-i\omega t}$; $k_P = \omega/\alpha$, $k_S = \omega/\beta$. Coordinates
 145 are ordered (z, x, y) with z down. The field at a voxel is the nine-component
 146 state

$$\psi = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \quad (1)$$

147 displacement followed by engineering strain. A voxel responds with a force
 148 and a moment (stress dipole); the 9×9 single-site T -matrix maps the exciting
 149 state to them, and the 9×9 lattice Green's tensor $\mathbf{K}(i, j)$ maps a source
 150 voxel's force and moment to the state at voxel i . The Foldy–Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j) \mathbf{T}_j \psi_j, \quad (2)$$

151 with ψ^0 the incident state.

152 *Where the kernel comes from.* Write the medium as the background plus
 153 the contrast. With time dependence $e^{-i\omega t}$ the equation of motion is $\partial_c \sigma_{bc} +$
 154 $\rho \omega^2 u_b = 0$. Splitting $\rho = \rho_0 + \Delta\rho$ and $\sigma = \mathcal{C}_0 : \varepsilon + \Delta\mathcal{C} : \varepsilon$ and moving the
 155 contrast terms to the right leaves the background operator on the left and
 156 the contrast on the right as a body force,

$$f_b = \omega^2 \Delta\rho u_b + \partial_c m_{bc}, \quad m_{bc} = (\Delta\mathcal{C} : \varepsilon)_{bc} : \quad (3)$$

157 a force from the density contrast, and the divergence of the contrast stress m ,
 158 the moment, from the stiffness contrast. With $\mathbf{G}$ the Green's tensor of the

background, the displacement is $u = u^0 + \int \mathbf{G} f \, d\mathbf{y}$, and the moment term
integrates by parts,

$$\int G_{ab}(\mathbf{x}-\mathbf{y}) \frac{\partial m_{bc}}{\partial y_c} \, d\mathbf{y} = - \int \frac{\partial G_{ab}}{\partial y_c}(\mathbf{x}-\mathbf{y}) m_{bc} \, d\mathbf{y} = + \int \frac{\partial G_{ab}}{\partial x_c}(\mathbf{x}-\mathbf{y}) m_{bc} \, d\mathbf{y}, \quad (4)$$

the boundary term vanishing because m is confined to the scatterers, and the
last step holding because $\mathbf{G}$ depends only on the separation. The moment
therefore radiates through the *receiver* derivative of $\mathbf{G}$, with a plus sign, when
the moment is the contrast stress itself. With the moment tensor defined
instead through its equivalent body force $-\partial_c[M_{bc}\delta(\mathbf{x}-\mathbf{y})]$, as in Aki and
Richards [24], the same field reads $M_{bc}\partial G_{ab}/\partial y_c$, the source derivative, with
 $M = -m$. The two are one convention written two ways; what matters is
that the derivative in the kernel and the sign in the contrast operator are
chosen together. The nine-component kernel follows: its force columns are
 $\mathbf{G}$ and its moment columns the receiver derivatives $\partial_c G_{ab}$, symmetrised over
 (b, c) with a factor $\frac{1}{2}$ on the shear components; its displacement rows are
those entries and its strain rows their receiver derivatives, symmetrised in
the same way and doubled on the shear rows to give engineering strain. In
that convention the contrast operator is

$$\Delta = \text{blockdiag}(\omega^2 \Delta \rho \mathbf{I}_3, \Delta \mathcal{C}), \quad (5)$$

where $\Delta \mathcal{C}$ is the 6×6 contrast stiffness with $\Delta \lambda + 2\Delta \mu$ and $\Delta \lambda$ in the normal
block and $2\Delta \mu$ (not $\Delta \mu$) on the shear diagonal.

Averaging, and collocation. Three operations in this paper are averages over
a cell, and they are distinct. The *source-cell average* integrates the kernel
over the cell that radiates: $\langle \mathbf{K} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{K}(\mathbf{r}-\mathbf{y}) \, d\mathbf{y}$ is the field at $\mathbf{r}$

180 of a uniform polarisation of the source cell. Every scheme in this paper uses
 181 it, and §5 shows it to be exact. *Collocation* fixes where the discrete equation
 182 is imposed: at the receiving cell's centre, with the internal field of each cell
 183 uniform, so that the unknown is the field at the centre. The *receiver-cell*
 184 *average* imposes the equation in the mean over the receiving cell instead;
 185 with a uniform internal field that is the Galerkin scheme of degree zero,
 186 called the mean-only voxel below. The two are both *uniform-field* voxels, a
 187 term that refers to the internal field and not to the test, and they differ only
 188 in the test function (a delta at the centre against the cell's indicator) and
 189 in the constant of their $O((kd)^2)$ error (Table 3). The *cell-mean contrast*,
 190 finally, is an average of the material, not of the field: where the medium
 191 varies within a cell, the contrast is projected onto the cell's basis, its mean
 192 for degree $r = 0$, rather than sampled at the centre (§§9, 10). In the language
 193 of the coupled-dipole method, the point dipole is collocation with the point
 194 kernel, the integrated Green tensor adds the source-cell average, and the
 195 filtered coupled dipoles replace the point source by one of finite extent.

196 *The laterally uniform layer.* A layer of thickness D is filled with n planes
 197 of voxels, $d = D/n$, and illuminated at normal incidence. The field is then
 198 laterally uniform, only the zero-lateral-wavenumber (specular) component of
 199 the lattice coupling acts, and Eq. (2) becomes exactly a chain of n planes
 200 coupled by the specular lattice sums

$$\mathbf{S}(m) = \sum_{\mathbf{R}} \mathbf{G}(\mathbf{R} + md\hat{\mathbf{z}}), \quad (6)$$

201 where the sum runs over the lattice vectors $\mathbf{R} = d(0, p, q)$, p, q integers, that
 202 join the centres of the voxels in one plane (a square lattice of pitch d), and

203 m is the separation of the two planes in units of d (Fig. 1a,b). The question
 204 is whether, and how fast, this chain converges to the continuous layer as
 205 $n \rightarrow \infty$ at fixed D .

206 *Implementations.* Every result is derived symbolically in Mathematica and
 207 compared with an independent numerical implementation or an exact solu-
 208 tion. “The package” below is the Python implementation of the voxel scheme
 209 (`cubic_scattering`): its cube T -matrix, its lattice kernels and its stratified
 210 propagator, which serve both as the object under test and, where validated,
 211 as a second implementation.

212 3. The continuous layer and its thin-layer expansion

213 At normal incidence a plane force drives a one-dimensional problem per
 214 wave type, with modulus M (M_P for P, μ for S), wavenumber k and plane
 215 Green’s function $g(z) = \frac{i}{2Mk} e^{ik|z|}$. The layer’s reflection and transmission
 216 follow from continuity of displacement and traction at its faces; they agree
 217 with the stratified propagator used as the reference in this work to 2.5×10^{-8} ,
 218 P and S (`ContinuumLimit_Reference.wl`).

219 For a layer of thickness D with modulus M_1 and density ρ_1 in a back-
 220 ground M_0, ρ_0 , write $Z_j = \sqrt{M_j \rho_j}$ for the impedances, $\zeta = Z_1/Z_0$ for their
 221 ratio and $k_1 = \omega \sqrt{\rho_1/M_1}$ for the layer’s wavenumber. The reflection coeffi-
 222 cient expands as

$$\begin{aligned}
 R = & \frac{ik_1 D}{2} (\zeta - \zeta^{-1}) + \frac{(k_1 D)^2}{4} (\zeta^{-2} - \zeta^2) - \frac{i(k_1 D)^3}{24} (\zeta - \zeta^{-1}) (3\zeta^{-2} + 2 + 3\zeta^2) \\
 & + \frac{(k_1 D)^4}{48} (\zeta^2 - \zeta^{-2}) (3\zeta^2 - 2 + 3\zeta^{-2}) + O(D^5),
 \end{aligned}
 \tag{7}$$

223 exact in the contrast (`ContinuumLimit_ThinLayer.wl`, which checks each
 224 coefficient symbolically and the truncated series against the exact layer).
 225 The first term is

$$\frac{i\omega D}{2} \sqrt{\frac{M_0}{\rho_0}} \left[\frac{\rho_1 - \rho_0}{M_0} - \rho_0 \left(\frac{1}{M_1} - \frac{1}{M_0} \right) \right] : \quad (8)$$

226 to first order a thin layer is an interface across which the traction jumps by
 227 the excess inertia and the displacement by the excess compliance. At oblique
 228 incidence the same statement holds with the first-order system $\partial_z \mathbf{b} = \mathbf{A} \mathbf{b}$
 229 for the P-SV state $\mathbf{b} = (u_x, u_z, t_{xz}, t_{zz})$: the layer acts to $O(D)$ as the jump
 230 $\mathbf{I} + (\mathbf{A}_1 - \mathbf{A}_0)D$, which reproduces the layer's $O(D)$ reflection coefficients
 231 R_{PP}, R_{PS} to 10^{-12} ; the series truncated at $O(D^k)$ converges at exactly order
 232 k ; and a soft thin layer with shear modulus $\mu_1 = D/\eta$ at fixed compliance η
 233 tends to the linear-slip interface as $D \rightarrow 0$. These expansions are the target
 234 a single plane of voxels must reproduce order by order. A plane of voxels
 235 that carry Legendre moments of their internal field to degree p reproduces
 236 Eq. (7) term by term through $O(D^{2p+2})$ and first differs at D^{2p+3} (§8): the
 237 mean-only voxel reproduces the equivalent interface and the D^2 term, the
 238 first-moment voxel all four terms shown.

239 4. The specular sums of the point Green's tensor

240 *Why transform the sum.* The specular sum (6) adds, at a receiver, the field
 241 of every voxel of a plane m planes away. In that form it is a poor object to
 242 study. The Green's tensor decays slowly with distance, so the sum converges
 243 slowly, and it hides the one question that matters here: which part of it is
 244 the continuum, and which part is the discretisation. The Poisson summation

245 formula [25] answers both at once, and is the standard tool for lattice sums
 246 of wave fields [26]. For a function f on the plane and the square lattice of
 247 pitch d it states

$$\sum_{\mathbf{R}} f(\mathbf{R}) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{f}(\mathbf{g}), \quad \hat{f}(\mathbf{g}) = \int f(\mathbf{x}) e^{-i\mathbf{g} \cdot \mathbf{x}} d^2x, \quad \mathbf{g} = \frac{2\pi}{d}(p, q), \quad (9)$$

248 with p, q integers: a sum of samples of f on the lattice equals a sum of samples
 249 of its Fourier transform on the reciprocal lattice, whose pitch is $2\pi/d$ (Fig. 2).
 250 Applied to the specular sum,

$$\mathbf{S}(m) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{\mathbf{G}}(\mathbf{g}; md), \quad (10)$$

251 with $\hat{\mathbf{G}}(\mathbf{g}; z)$ the two-dimensional (lateral) transform of the Green's tensor
 252 between planes a distance z apart. Each term has a physical meaning. Trans-
 253 formed laterally, the field of a plane of sources is a superposition of plane
 254 waves, one for each lateral wavenumber, and the lattice selects the wavenum-
 255 bers $\mathbf{g}$. The order $\mathbf{g} = 0$ is the laterally uniform plane wave that a continuous
 256 layer radiates: it is the continuum's term. Every other order has $|\mathbf{g}| \geq 2\pi/d$,
 257 far beyond the wavenumber of the field when the voxels are small compared
 258 with the wavelength, so it is evanescent and decays away from the plane as
 259 $e^{-|\mathbf{g}|md} = e^{-2\pi m \sqrt{p^2+q^2}}$, independently of d . The transform therefore splits
 260 the lattice sum exactly into the continuum and a discretisation correction,
 261 and the correction converges after a few orders.

262 *The local term that survives.* In the strain-from-moment block the evanescent
 263 orders carry two powers of $|\mathbf{g}| \propto 1/d$, so their contribution per voxel volume
 264 does *not* vanish as $d \rightarrow 0$: at $m = 1$ it is eleven times the continuum

265 term (`ContinuumLimit_SpecularSums.wl`, matched to the package's Ewald
 266 lattice sum to 5×10^{-13}). This surviving term is static. Summed over all
 267 planes in slab order, the static lattice sum of the point kernel is exactly
 268 the depolarisation of a uniform plate ($-1/M_P$ on $\varepsilon_{zz} \leftarrow M_{zz}$ and $-1/(2\mu)$
 269 on the normal shears, per unit volume) plus a cubic-symmetric remainder
 270 (`ContinuumLimit_StaticLocal.wl`, to 2×10^{-10}). The point kernel therefore
 271 reproduces the continuum only if that remainder is cancelled by the voxel's
 272 own self term.

273 5. The tiling identity

274 Replace the point coupling by the *source-cell average* $\langle \mathbf{G} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{G}(\mathbf{r}-$
 275 $\mathbf{y}) d\mathbf{y}$, the field at a voxel centre of a uniform polarisation of the source voxel.
 276 Each Poisson order in Eq. (10) is then multiplied by the cube's form factor

$$F(\mathbf{g}) = \text{sinc}(g_x h) \text{sinc}(g_y h), \quad (11)$$

277 which vanishes at every $\mathbf{g} \neq 0$ of the reciprocal lattice since $\text{sinc}(p\pi) = 0$ for
 278 integer $p \neq 0$. This is the Fourier form of the fact that cubes tile the plane,
 279 $\sum_{\mathbf{R}} \chi(\mathbf{x} - \mathbf{R}) = 1$ for the cube's indicator χ : a partition of unity (Fig. 3).
 280 Two exact statements follow.

281 *Between planes.* For $m \neq 0$ every evanescent order is removed and

$$\mathbf{S}_{\text{avg}}(m) = \frac{1}{d^2} \sum_{W=P,S} \hat{\mathbf{G}}_W(0; md) \text{sinc}(k_W h), \quad (12)$$

282 the continuum's plane-wave term times the vertical form factor. The pack-
 283 age's cell-averaged kernel agrees in all 81 entries to 2.8×10^{-16} (`ContinuumLimit_`
 284 `AveragedSums.wl`).

285 *Within a plane.* The sum over *all* cells of the plane, the voxel's own included,
 286 is the field at the mid-plane of a uniform plate of thickness d (Fig. 1c):

$$d^3 \mathbf{S}_{\text{avg}}(0) + \mathbf{S}_{\text{self}} = d^3 \mathbf{P}_0, \quad \mathbf{S}_{\text{self}} = \int_{\text{cube}} \mathbf{G}(-\mathbf{y}) d\mathbf{y}, \quad (13)$$

287 where $\mathbf{P}_0 = d^{-3} \int_{-h}^h \mathbf{G}_{\text{plate}}(-z') dz'$ is elementary: in the one-dimensional
 288 transform each entry is $c_0 + A/(Mk_z^2 - \rho\omega^2)$, so it averages to $c_0/d + A(e^{ikh} -$
 289 $1)/(dMk^2)$. Its only nonzero entries are $u \leftarrow f$ on the diagonal and the
 290 three normal strain-from-moment entries, whose local weights $-1/M_P$ and
 291 $-1/(2\mu)$ are the plate depolarisation of §4. Statically the identity holds for
 292 all 21 independent components of the strain-from-moment tensor to 4×10^{-9} ,
 293 with every cell integral evaluated analytically (below).

294 *Consequence: an exact kernel with no lattice sum.* At normal incidence the
 295 same-plane block of the cell-averaged Foldy-Lax kernel is

$$\mathbf{S}_{\text{avg}}(0) = \mathbf{P}_0 - \mathbf{S}_{\text{self}}/d^3, \quad (14)$$

296 with no Ewald splitting, near shell or asymptotic tail. The route it replaces
 297 (Ewald summation with a near shell averaged by quadrature and an $O(d^2)$
 298 Taylor tail beyond it) carries a static strain-from-moment error of 9.4×10^{-4}
 299 at its default settings, falling to 2.6×10^{-5} only when both the shell radius
 300 and the quadrature order are raised; Eq. (14) is now the package's kernel at
 301 normal incidence.

302 *Cell integrals without quadrature.* Write the static (Kelvin) tensor as $G_{bc} =$
 303 $(c_A + c_B) \delta_{bc}/r - c_B \partial_b \partial_c r$ with $c_B = (1/\mu - 1/M_P)/8\pi$ and $c_A = 1/(4\pi\mu) - c_B$.
 304 Every cell integral of its derivatives is a cell integral of derivatives of $1/r$ and
 305 r . One application of the divergence theorem moves it onto faces $x_p = c_p \pm h$,

306 which never contain the origin, and there the two-dimensional antiderivative
 307 in the face coordinates is continuous, so the face integral is a sum over the
 308 face's corners. The distributional term at $r = 0$ is carried by the divergence
 309 theorem itself. (A triple antiderivative evaluated at the cube's corners is
 310 *not* usable for repeated derivatives: its arctangent terms jump across the
 311 coordinate planes through the origin, and differentiating twice there discards
 312 exactly that term.) The self term so obtained reproduces the classical closed
 313 form of the cube's depolarisation to machine precision.

314 **6. The single site: the cube T -matrix from moment integrals**

315 Every error of a voxel scheme is made at the single site (§11), so the single
 316 site is derived here rather than imported. The classical route imports it. For
 317 an ellipsoid, Eshelby's uniformity theorem makes the internal strain under a
 318 uniform load uniform, with a depolarisation tensor in closed form [23, 27, 28];
 319 its dynamic generalisation is known for spheres and ellipsoids [29, 30]. For
 320 a cube or a general polyhedron the static field of a uniform eigenstrain is
 321 also in closed form [31, 32, 33, 34], but the strain it produces is not uniform,
 322 so there is no exact uniform-strain inclusion to import. In scattering theory
 323 the same object appears as the self-consistent or effective-field closure of
 324 the volume integral equation [35, 16, 36]: a first-order (Born) scatterer [37]
 325 renormalised by the geometric series of its own near field, the step that turns
 326 a contrast into an effective contrast. Nestor [17] introduced the single site
 327 in this form, summing the series of the cell's self integral, with the singular
 328 near field evaluated over an infinitesimal cell. The moments below evaluate
 329 the same self integral over the finite cube, as a distribution. The elastic T -

matrix of Waterman [38] represents a scatterer in spherical multipoles, the natural basis for a sphere; a voxel in a lattice needs one compatible with the lattice, and the moments over the cube provide it. The construction below makes the approximation of a voxel explicit rather than implicit: it derives the cube's T -matrix from the moments of the elastodynamic Green's tensor over the cube, and identifies the uniform-strain closure as one truncation of an exact hierarchy.

6.1. The closed hierarchy

Let the voxel be $V = [-h, h]^3$ with contrast $\delta c_{ijkl} = \Delta\lambda \delta_{ij}\delta_{kl} + \Delta\mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$ and $\Delta\rho$. Expand the internal displacement about the centre to second degree and project the Lippmann–Schwinger equation onto 1, x_p and x_px_q . The unknowns are $u_i(0)$, $\partial_p u_i(0)$ and $\partial_p \partial_q u_i(0)$, and the result is the closed system

$$(\delta_{ij} - \omega^2 \Delta\rho \mathbf{G}_{ij}) u_j + (\mathbf{N}_{in,k}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta\rho \mathbf{K}_{ij}^{rs}) \partial_r \partial_s u_j = u_i^0, \quad (15)$$

$$(\delta_{pr} \delta_{ij} + \omega^2 \Delta\rho \mathbf{N}_{ij,p}^r - \mathbf{M}_{in,pk} \delta c_{nkrj}) \partial_r u_j = \partial_p u_i^0, \quad (16)$$

$$-\omega^2 \Delta\rho \mathbf{M}_{ij,pq} u_j + (\delta_{qr} \delta_{ps} \delta_{ij} + \mathbf{Q}_{in,pqk}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta\rho \mathbf{P}_{ij,pq}^{rs}) \partial_r \partial_s u_j = \partial_p \partial_q u_i^0, \quad (17)$$

all fields evaluated at the centre. Its coefficients are six moments of the background Green's tensor $\mathbf{G}$ over the voxel,

$$\begin{aligned} \mathbf{G}_{ij} &= \int_V G_{ij} \, dV, & \mathbf{K}_{ij}^{rs} &= \int_V G_{ij} x_r x_s \, dV, & \mathbf{N}_{in,k}^r &= \int_V \partial'_k G_{in} x_r \, dV, \\ \mathbf{M}_{in,pk} &= \int_V \partial'_p \partial'_k G_{in} \, dV, & \mathbf{P}_{ij,pq}^{rs} &= \int_V \partial'_q \partial'_p G_{ij} x_r x_s \, dV, & \mathbf{Q}_{in,pqk}^r &= \int_V \partial'_q \partial'_p \partial'_k G_{in} x_r \, dV, \end{aligned} \quad (18)$$

345 with derivatives taken with respect to the source point, $\partial' = -\partial$. That sign
 346 reaches **N** and **Q** only, and no symmetry, parity or scaling check can detect
 347 it, so it is fixed by convention. The moments are graded by (D, W) , the
 348 number of derivatives and the degree of the weight: **G** $(0, 0)$, **K** $(0, 2)$, **N** $(1, 1)$,
 349 **M** $(2, 0)$, **P** $(2, 2)$, **Q** $(3, 1)$.

350 6.2. The moments as distributions

351 Write the elastodynamic tensor as

$$G_{in}(\mathbf{r}) = \frac{1}{4\pi\mu} \left[\delta_{in} g_S(r) + \frac{1}{k_S^2} \partial_i \partial_n (g_S(r) - g_P(r)) \right], \quad g_c(r) = \frac{e^{ik_c r}}{r} \quad (19)$$

352 [39, 24]. Expanding g_c in powers of r reduces every moment in Eq. (18) to a
 353 finite sum of scalar moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \rangle. \quad (20)$$

354 For $m = -1$ and $D \geq 2$ the classical integrand is not absolutely integrable.
 355 $\partial^2(1/r)$ decays as r^{-3} , its angular average vanishes, and the value of the inte-
 356 gral depends on the shape of the region excluded around the origin. This is
 357 the source of the δ -function term of the inclusion problem, which a numerical
 358 quadrature misses. Without it the depolarisation is wrong; for a sphere it
 359 vanishes altogether (§6.3). It is not added afterwards here. The derivatives
 360 are distributional from the start, and each is moved onto the voxel's faces by
 361 pairing with the test function $w \mathbf{1}_V$, using $\partial_q \mathbf{1}_V = -n_q \delta_{\partial V}$:

$$\langle \partial_q T, w \mathbf{1}_V \rangle = \oint_{\partial V} n_q w T \, dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (21)$$

362 The surface term is evaluated on the faces, a distance h from the origin, where
 363 every derivative of r^m is smooth. The recursion lowers D and W together

and ends on convergent integrals. The face integrals are done in closed form by the corner sums of §5. The sharp check is the sum rule

$$\sum_p E[-1; p, p;] = \nabla^2(1/r) \text{ paired with } \mathbf{1}_V = -4\pi, \quad (22)$$

which the face route satisfies exactly and a spherical excision returns as 0. The closed forms contain logarithms of the units $(2 \pm \sqrt{3})^n$ of the Pell equation $x^2 - 3y^2 = 1$. Left unreduced, these cancel catastrophically: for example, $\log(72010600134783751 - 41575339372323900\sqrt{3})$ is a cancellation of 35 digits. They are therefore reduced to $n \log(2 - \sqrt{3})$, which is exact. Because every power of r is paired this way, the dynamic moments keep the even-power terms of the expansion $(k^2 r, r^3, \dots)$ as well as the odd ones. A route that integrates only the polynomial (odd-power) terms of $\mathbf{G}$ misses those terms, at a real $O((kh)^2)$ cost (§6.4).

6.3. Parity, and the nine-component block

The cube is centrosymmetric. A moment that couples a field quantity of degree a to one of degree b therefore vanishes unless $a + b$ is even. u , ∂u and $\partial\partial u$ have degrees 0, 1 and 2, so Eq. (16) contains the first gradient alone: ∂u is determined by its own block and couples to neither u nor $\partial\partial u$. The leading scattering by a stiffness contrast is the stress dipole $\int_V \delta c : \nabla u \, dV$, which needs only ∂u . The leading far field of the stiffness contrast is therefore fixed by Eq. (16), and *no treatment of the second-gradient block (17) can change it*. What Eq. (17) contributes reaches the far field only through the $O((kh)^2)$ coupling in Eq. (15). That is the order, and the origin (the variation of the field across the voxel), of the collocation error of §7, which the first-moment voxel of §8 removes.

387 With $\Delta\rho = 0$, Eq. (16) is the 9×9 operator $\delta_{pr}\delta_{ij} - \mathbf{M}_{in,pk}\delta c_{nkrj}$, built
 388 from $\mathbf{M}$ alone. Under O_h the nine gradient components decompose as $A_{1g} \oplus$
 389 $E_g \oplus T_{1g} \oplus T_{2g}$, and the operator is diagonal in that basis. Its eigenvalues are
 390 the inverse strain concentrations:

$$A_{1g} \text{ (dilatation)} : 1 + \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{\lambda + 2\mu}, \quad (23)$$

$$T_{2g} \text{ (off-diagonal shear)} : 1 + 2\Delta\mu S_{\text{off}}, \quad S_{\text{off}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (24)$$

$$E_g \text{ (diagonal shear)} : 1 + 2\Delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\pi\mu}{6\pi\mu(\lambda + 2\mu)}. \quad (25)$$

391 The density channel is the displacement block of Eq. (15) with the second
 392 gradients dropped, an amplification $(1 - \omega^2\Delta\rho\mathbf{G})^{-1}$ with $\mathbf{G}$ the cube integral
 393 of $\mathbf{G}$.

394 *The cube against the sphere.* For a sphere, Eshelby's tensor gives a dilata-
 395 tional channel $1 + \Delta K/(\lambda + 2\mu)$, with $\Delta K = \Delta\lambda + \frac{2}{3}\Delta\mu$, and a single deviatoric
 396 channel $1 + 2\Delta\mu S_{\text{sph}}$, with $2\mu S_{\text{sph}} = 2(4 - 5\nu)/[15(1 - \nu)]$. The cube's di-
 397 latational channel, Eq. (23), is the sphere's exactly: its eigenvalue involves
 398 $\mathbf{M}$ only through the contraction $\int_V \partial_i \partial_n G_{in} dV$, and the static Navier equa-
 399 tion makes $\partial_i \partial_n G_{in} = -\delta(\mathbf{r})/(\lambda + 2\mu)$, whose integral does not depend on
 400 the shape. The cube's shear channels differ from each other. Their isotropic
 401 average, however, is the sphere's, also exactly:

$$\frac{1}{5}(3S_{\text{off}} + 2S_{\text{diag}}) = S_{\text{sph}}, \quad S_{\text{diag}} - S_{\text{off}} = \frac{(\lambda + \mu)(5\sqrt{3} - 2\pi)}{6\pi\mu(\lambda + 2\mu)} > 0. \quad (26)$$

402 The $\sqrt{3}$ terms cancel in the average. What a cube adds to a sphere at this
 403 order is therefore a pure cubic anisotropy. It splits the deviatoric channel

about the sphere's value, weighted 3 : 2 by the dimensions of T_{2g} and E_g ,
and the diagonal shear is the more depolarised. At the background of §7
($\nu = 7/32$), $2\mu S_{\text{off}} = 0.4314$, $2\mu S_{\text{diag}} = 0.5929$ and the sphere's value is
0.4960.

Where Eshelby enters. Equations (23)–(25) follow from the moment integrals
and the group theory alone. No ellipsoidal result is used, and no δ term is
added by hand. What corresponds to Eshelby is the *truncation*. Keeping
 ∂u and discarding $\partial\partial u$ is exactly the uniform-internal-strain ansatz, and by
the uniformity theorem that ansatz is exact if and only if the inclusion is
an ellipsoid. For a sphere the construction is therefore exact in the static
limit, and for a cube it is an approximation. The hierarchy (15)–(17) states
that approximation as the omitted blocks, rather than as an unquantified
shape correction. A polynomial expansion about the centre cannot represent
the edge and corner concentrations of an isolated cube's internal field at any
finite degree. A voxel in a uniform layer has no such concentrations: its
neighbours have the same properties, so its internal field is the continuum's,
which is smooth. That is why a low-degree internal field suffices in a lattice
(§§7–8) although it is only an approximation for an isolated cube.

Against the exact sphere. These statements are tested against the exact elas-
tic Mie solution for a sphere, with no Eshelby formula in between. The
concentration of Mie order n is $E_n = a_n/a_n^{\text{Born}}$ (orders 0, 1, 2 are the dilata-
tional, density and shear channels), extrapolated to $ka \rightarrow 0$. The formulation
is applied to a ball of radius a , with its moment $\mathbf{M}$ evaluated by the surface
route of Eq. (21) on $|\mathbf{r}| = a$. Its dilatational and shear channels equal Mie's
to 2×10^{-10} , and the density channel is 1. The cube's closed forms meet

429 the same numbers. Its dilatational channel is Mie's E_0 to 2×10^{-10} , and its
 430 isotropic shear average, Eq. (26), is the depolarisation read from Mie's E_2
 431 to 10^{-8} . The same moment taken as a principal value, over the ball with
 432 a small sphere excluded around the origin, has zero depolarisation in every
 433 channel. The whole static response of the sphere is the δ term.

434 At finite frequency the closure uses the dynamic ball moments, which are
 435 in closed form. It departs from Mie at second order in ka and first order in
 436 the contrast, and the departure is closed-form too: the ball's moments are
 437 elementary, and Mie's low-frequency series with the contrast kept exact gives
 438 E_n to relative order $(ka)^3$. To first order in the contrast,

$$\frac{E_n^{\text{closure}}}{E_n^{\text{Mie}}} - 1 = \begin{cases} -\frac{(k_P a)^2}{10} \frac{\Delta \lambda}{\lambda + 2\mu}, & n = 0, \\ \frac{2(k_S a)^2 + (k_P a)^2}{30} \frac{\Delta \rho}{\rho}, & n = 1, \\ -\frac{3(k_S a)^2 + 2(k_P a)^2 \beta^2 / \alpha^2}{75} \frac{\Delta \mu}{\mu}, & n = 2, \end{cases} \quad (27)$$

439 with no $O(ka)$ term and no $O((ka)^3)$ term. The imaginary $(ka)^3$ terms,
 440 the radiation damping, agree exactly in all three channels. So the closure's
 441 only error at this order is the real $(ka)^2$ term, the uniform-field ansatz's
 442 own. That is the order the hierarchy assigns to the omitted second gradi-
 443 ents (§6.3), and it has the same origin, the variation of the field across the
 444 scatterer, as the collocation error of §7. Eq. (27) is derived symbolically
 445 (`SphereClosureDynamic.wl`). It agrees with the coefficient extracted from
 446 the numerical Mie solve and an independent implementation of the closure
 447 to 5×10^{-4} , which is the size of the next order in the contrast.

448 6.4. The collocation closure

449 The collocation closure takes the internal field of a voxel as uniform, the
 450 truncation of §6.3 for the density and stiffness channels together. The voxel's
 451 own field at its centre is then $\mathbf{S}_{\text{self}}\Delta e$ for internal state e , where $\mathbf{S}_{\text{self}}$ is the
 452 nine-component form of the moments $\mathbf{G}$ and $\mathbf{M}$. By parity it is block-diagonal.
 453 So $e = \psi + \mathbf{S}_{\text{self}}\Delta e$ and

$$\mathbf{T} = d^3 \Delta (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)^{-1}. \quad (28)$$

454 With the kernel (14) the self term cancels. Substituting $\psi = (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)e$
 455 into Eq. (2),

$$e_i = \psi_i^0 + d^3 \sum_j \mathbf{K}_{\text{all}}(i, j) \Delta e_j, \quad \mathbf{K}_{\text{all}}(i, i) = \mathbf{P}_0, \quad \mathbf{K}_{\text{all}}(i, j) = \mathbf{S}_{\text{avg}}(i - j), \quad (29)$$

456 an identity checked on the matrices to 4×10^{-16} . Every block of $\mathbf{K}_{\text{all}}$ is an
 457 *exact* cell integral of the one-dimensional continuum kernel (Eqs. (12), (13)),
 458 so Eq. (29) is precisely the collocation of the continuum Lippmann–Schwinger
 459 equation with a strain piecewise constant over voxels. It does not contain
 460 $\mathbf{S}_{\text{self}}$ at all.

461 The package's cube T -matrix is this closure statically: both blocks agree
 462 with Eq. (28) to 2×10^{-5} at $k_S h = 0.01$, the difference falling as $(kh)^2$
 463 (`ContinuumLimit_CubeT.w1`). It departs dynamically by about $0.25(kh)^2$
 464 through its far-field form-factor corrections. Its self integrals are exact in the
 465 static and radiation parts but omit the real even-power terms of the Green's
 466 tensor's expansion ($k^2 r, r^3, \dots$), an error of 0.4–0.6 $(kh)^2$ against the analytic
 467 self term, whose series agrees with the closed-form elastodynamic tensor to
 468 10^{-14} . By Eq. (29) that error cannot enter the closure scheme.

469 7. Convergence, and the closed-form error

470 *Second order.* For a layer $D = 2\text{ m}$ with the contrast $\Delta\lambda = 2\text{ GPa}$, $\Delta\mu =$
 471 1 GPa , $\Delta\rho = 100\text{ kg m}^{-3}$ in a background $\alpha = 5000$, $\beta = 3000\text{ m s}^{-1}$, $\rho =$
 472 2500 kg m^{-3} , illuminated by a P plane force, the reflected displacement of
 473 the chain converges to the continuous layer's at order 2.000 for $n = 4 \rightarrow$
 474 16 at $\omega = 60, 300$ and 600 rad s^{-1} , with the collocation (29) and with the
 475 package *T*-matrix alike (`ContinuumLimit_Chain.wl`). The error depends on
 476 $k_S h$ alone. Throughout, an order quoted for a pair of grids is the apparent
 477 order $\ln(e_1/e_2)/\ln(n_2/n_1)$ between them, with e the relative error against
 478 the exact solution and n the refinement; the orders in the legend of Fig. 4
 479 are least-squares slopes of $\ln e$ against $\ln n$ over the grids marked as fitted,
 480 with their standard errors. The two agree where a sequence is asymptotic
 481 and differ where it is not, which the apparent orders show.

482 *The leading error in closed form.* In Eq. (29) the kernel is integrated exactly;
 483 the only approximation is the internal field's variation across a voxel. At first
 484 order in the contrast the discrete scattered field is therefore a midpoint rule
 485 applied to the continuum's Born integral, with the kernel's cell integral exact.
 486 For reflection the integrand carries the two-way phase e^{2ikz} ; for transmission
 487 it is flat. With k the background wavenumber of the wave concerned (k_P for
 488 an incident P wave, k_S for an incident S wave), R the reflection coefficient
 489 and T^{sc} the scattered part of the transmitted field, for *every* n

$$\frac{R_{\text{disc}}}{R} = \frac{\text{sinc}(kh)}{\text{sinc}(2kh)} = 1 + \frac{(kd)^2}{8} + O(d^4), \quad \frac{T_{\text{disc}}^{\text{sc}}}{T^{\text{sc}}} = \text{sinc}(kh) = 1 - \frac{(kd)^2}{24} + O(d^4), \quad (30)$$

490 confirmed for $n = 1$ to 16 to 10^{-4} of the error itself (`ContinuumLimit_`
 491 `ErrorConstant.wl`); for an incident S wave, with $k = k_S$, the same forms
 492 and those of Eq. (33) hold to 4×10^{-4} (`ContinuumLimit_IncidentS.wl`, §8).
 493 At the full contrast the reflection constant is unchanged (0.124 against 1/8);
 494 the transmission constant grows to -0.092 .

495 *Why a uniform-field single site stops at second order.* The two constants have
 496 opposite signs, so no scalar correction of $\mathbf{T}$, no single $O((kh)^2)$ factor, can
 497 cancel both. The package *T*-matrix illustrates it: its form-factor corrections
 498 reduce the reflection error by a factor 4.45 and *increase* the transmission
 499 error by 1.7, at every n . The missing term in Eq. (30) is the first moment
 500 of the internal field over each voxel, which radiates through the derivative of
 501 the kernel. A fourth-order scheme must carry it: a single site with a gradient
 502 channel, and a kernel whose cell integrals include the kernel's derivative.

503 8. Fourth order and beyond: Legendre moments of the voxel field

504 Let each voxel carry the mean *and* the first moment of its internal field.
 505 On voxel j , with local coordinate $s \in [-h, h]$ and Legendre functions $\phi_b(s) =$
 506 $P_b(s/h)$ ($\phi_0 = 1$, $\phi_1 = s/h$, $\phi_2 = (3s^2/h^2 - 1)/2$, ...), the field is $w =$
 507 $\sum_b c_{b,j} \phi_b$, $w = (u, \varepsilon)$ the displacement and strain of the P channel, and the
 508 polarisation is $q = (\omega^2 \Delta \rho u, \Delta M \varepsilon)$ with $\Delta M = \Delta \lambda + 2\Delta \mu$. Testing the
 509 continuum equation

$$w(z) = w^0(z) + \int \mathbf{K}(z-z') q(z') dz', \quad \mathbf{K} = \begin{pmatrix} g & g' \\ g' & g'' \end{pmatrix}, \quad g'' = -k^2 g - \delta(z)/M_P, \quad (31)$$

510 with the same functions on every voxel gives

$$\left(\int_i \phi_a^2 dz\right) c_{a,i} = \int_i \phi_a w^0 dz + \sum_{j,b} \left[\iint_j \phi_a(z) \mathbf{K}(z-z') \phi_b(z') dz dz' \right] \Delta_1 c_{b,j}, \quad (32)$$

511 $\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$. Between voxels the double integral is separable, a
 512 product of Legendre moments of $e^{\pm i k z}$; within a voxel it is split at $z' = z$
 513 and the local term contributes $-\int_i \phi_a \phi_b dz / M_P$. Every entry is closed-form.
 514 Keeping only ϕ_0 recovers a second-order scheme; keeping ϕ_0 and ϕ_1 con-
 515 verges at fourth order in both reflection and transmission (`ContinuumLimit_`
`FourthOrder.wl`; Table 1). A single plane of first-moment voxels represents

Table 1: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, against the number of voxel planes n : collocation, the mean-only voxel and the first-moment voxel of Eq. (32). The order is 4.00 at $\omega = 60$ and 600 rad s^{-1} too.

n	collocation	mean only	mean and first moment
1	$1.8 \times 10^{-3} / 1.3 \times 10^{-3}$	$1.2 \times 10^{-3} / 1.8 \times 10^{-3}$	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$
4	$1.1 \times 10^{-4} / 8.3 \times 10^{-5}$	$7.5 \times 10^{-5} / 1.1 \times 10^{-4}$	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$
16	$7.0 \times 10^{-6} / 5.2 \times 10^{-6}$	$4.7 \times 10^{-6} / 6.9 \times 10^{-6}$	$4.2 \times 10^{-12} / 4.2 \times 10^{-12}$
order	2 / 2	2 / 2	4 / 4

516

517 the 2 m layer 6500 times more accurately than a single plane of uniform ones.
 518 The collocation column is the scheme of §6 and reproduces the chain of the
 519 full 9×9 system to all printed digits.

520 *Higher moments.* Carrying the Legendre moments to degree p changes noth-
 521 ing in the construction: the moments and the same-cell double integrals of
 522 $P_b(s/h)$ against exponentials are again closed-form, and for $p \leq 1$ the general-

523 degree solver reproduces the chain above exactly. The second-moment voxel
 524 ($p = 2$) converges at sixth order and the third-moment voxel ($p = 3$) at
 525 eighth, in reflection and transmission (Table 2; `ContinuumLimit_SecondMoment_`
 526 `w1`); the second-moment voxel's order is 6.00 at $\omega = 60, 300$ and 600 rad s^{-1} .
 The errors fall to 10^{-22} , so they are computed in 40-digit arithmetic.

Table 2: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, for voxels carrying Legendre moments to degree p . The order is the mean over the halvings $n = 2 \rightarrow 4 \rightarrow 8$.

n	first moment ($p = 1$)	second moment ($p = 2$)	third moment ($p = 3$)
1	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$	$2.8 \times 10^{-11} / 4.2 \times 10^{-11}$	$1.5 \times 10^{-15} / 1.5 \times 10^{-15}$
2	$1.7 \times 10^{-8} / 1.7 \times 10^{-8}$	$4.4 \times 10^{-13} / 6.5 \times 10^{-13}$	$6.0 \times 10^{-18} / 6.0 \times 10^{-18}$
4	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$	$6.9 \times 10^{-15} / 1.0 \times 10^{-14}$	$2.4 \times 10^{-20} / 2.4 \times 10^{-20}$
8	$6.7 \times 10^{-11} / 6.7 \times 10^{-11}$	$1.1 \times 10^{-16} / 1.6 \times 10^{-16}$	$9.2 \times 10^{-23} / 9.2 \times 10^{-23}$
order	4 / 4	6 / 6	8 / 8

527

528 *The leading error in closed form.* Without scattering, the Galerkin solution
 529 is the L^2 projection Π of the incident field onto each voxel's basis, and the
 530 kernel is integrated over each voxel exactly; at first order in the contrast the
 531 error is therefore the kernel's cell integral applied to $\Pi w^0 - w^0$. The field
 532 varies as e^{iks} across a voxel and the kernel toward an outside observer as e^{iqs} ,
 533 $q = +k$ for reflection and $q = -k$ for transmission, so for every scheme and
 534 every n

$$\frac{\text{discrete}}{\text{exact}} = \frac{\int_{-h}^h e^{iqs} \Pi[e^{iks}] \, ds}{\int_{-h}^h e^{i(q+k)s} \, ds}, \quad (33)$$

535 with Π the centre value for collocation, the cell mean for the mean-only voxel
 536 and the projection onto $\{P_0, \dots, P_p\}$ for the voxel of degree p (`ContinuumLimit_`

537 `FourthOrderError.wl`, `ContinuumLimit_SecondMoment.wl`). The results
 are in Table 3. The chain of Eq. (32) reproduces the forms for collocation

Table 3: The leading error of the discrete layer in closed form, Eq. (33): discrete over
 exact scattered field at first order in the contrast, for every number of voxel planes n .

	reflection	transmission
collocation	$1 + (kd)^2/8 + O(d^4)$	$1 - (kd)^2/24 + O(d^4)$
mean only	$1 + (kd)^2/12 + O(d^4)$	$1 - (kd)^2/12 + O(d^4)$
first moment	$1 - (kd)^4/720 + O(d^6)$	$1 - (kd)^4/720 + O(d^6)$
second moment	$1 + (kd)^6/100800 + O(d^8)$	$1 - (kd)^6/100800 + O(d^8)$
third moment	$1 - (kd)^8/25401600 + O(d^{10})$	$1 - (kd)^8/25401600 + O(d^{10})$
degree p	$1 + (-1)^p c_p (kd)^{2p+2}$	$1 - c_p (kd)^{2p+2}$

538
 539 and degrees 0 and 1 at $n = 1$ to 8 to 10^{-4} of the error, the remainder being
 540 the second order in the contrast, and those for degrees 2 and 3 to 4×10^{-7}
 541 at a contrast reduced by 10^{-6} . The constant for degree p follows from the
 542 Legendre expansion $e^{iks} = \sum_j i^j (2j+1) j_j(kh) P_j(s/h)$. The projection drops
 543 the terms $j > p$, led by $j = p+1$, whose coefficient is $O((kh)^{p+1})$; its moment
 544 against e^{iqs} is $2h i^{p+1} j_{p+1}(qh) = O((qh)^{p+1})$. With $j_n(x) \approx x^n/(2n+1)!!$ and
 545 $h = d/2$,

$$c_p = \frac{2p+3}{4^{p+1} [(2p+3)!!]^2} = \frac{[(p+1)!]^2}{(2p+2)! (2p+3)!}, \quad c_0 = \frac{1}{12}, \quad c_1 = \frac{1}{720}, \quad c_2 = \frac{1}{100800},$$

(34)

546 and the leading error carries $(q/k)^{p+1}$: the same sign in reflection and trans-
 547 mission for odd p , opposite signs for even p . The first-moment voxel's leading
 548 error therefore does not depend on the direction in which the field is observed.

Eq. (34) is checked symbolically for $p = 0$ to 5 (every lower power of kd vanishes) in Mathematica and, independently, in sympy.

One plane of voxels and the thin-layer series. Refining at fixed thickness is one limit; thinning a single plane is the other. For one plane of voxels ($n = 1$, $d = D$) the series in D of the reflection coefficient, computed exactly at the full contrast, coincides with that of the exact layer, Eq. (7), term by term through D^{2p+2} and first differs at D^{2p+3} , for $p = 0$ to 3 and for a layer both stiffer and softer than its surroundings (`ContinuumLimit_SecondMoment.w1`). Each further Legendre moment adds two terms of the thin-layer series.

In three dimensions. The three-dimensional first-moment voxel carries the nine-component state with Legendre weights $\{1, x/h, y/h, z/h\}$, 36 unknowns in all, tested with the same functions and coupled by cell-to-cell double integrals of the Green's tensor. For a laterally uniform layer at normal incidence it reduces *exactly* to Eq. (32) (`ContinuumLimit_Reduction3D.w1`). The lateral first moments are never excited: the lattice and the incident field are symmetric under reflection about every voxel's centre planes, and $x/h, y/h$ are odd there. The mean and the z -moment carry weights that are constant laterally, so their lateral form factor $\text{sinc}(g_x h) \text{sinc}(g_y h)$ vanishes on the reciprocal lattice and the plane sum of the double integrals is the plate-to-plate integral in z alone. Two facts mark where genuinely three-dimensional structure begins. The lateral first moment's form factor, $F_1(g) = i(gh \cos gh - \sin gh)/(gh)^2$, does not vanish on the reciprocal lattice ($F_1 = i(-1)^p/p\pi$), so the gradient channels couple through the evanescent orders. And at oblique incidence the mean's own zeros shift,

573 $\text{sinc}((2\pi p/d + k_x)h) = (-1)^p k_x h / p\pi + O(k_x^2 h^2)$, so the tiling identity holds
574 only to $O(k_{\parallel} h)$.

575 *Oblique incidence.* The scheme is built spectrally. By Poisson summation
576 the Bloch coupling between planes i and j at lateral wavenumber $\mathbf{k}_{\parallel}$ is

$$C_{ab}(i, j) = \frac{1}{d^2} \sum_{\mathbf{g}} L_a(\boldsymbol{\kappa}) L_b(-\boldsymbol{\kappa}) \iint \phi_a(z) \hat{\mathbf{G}}(\boldsymbol{\kappa}; z - z') \phi_b(z') dz dz', \quad \boldsymbol{\kappa} = \mathbf{k}_{\parallel} + \mathbf{g}, \quad (35)$$

577 with L_a the lateral Legendre factors. The whole-space tensor decays as
578 $1/k_z^2$ and the nine-component state adds at most two powers of k_z , so each
579 entry of $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z)$ is a $\delta(z)$ of constant weight plus, per mode, $[\mathrm{i}\alpha/2\gamma +$
580 $\mathrm{i}\beta \text{sgn}(z)/2] \mathrm{e}^{\mathrm{i}\gamma|z|}$, $\gamma^2 = k_W^2 - \kappa^2$: every z integral is closed-form. Double
581 averaging makes the lattice sum converge without Ewald splitting, to three
582 digits at $|p|, |q| \leq 8$. Against the exact layer, for a P wave incident at 20°
and 40° (`ContinuumLimit_Oblique.wl`; Table 4). The mean-only voxel is

Table 4: Relative error of the reflected P (R_{PP}) and converted S (R_{PS}) waves for a P wave incident at 20° , with the mean-only and first-moment voxels of Eq. (35). At 40° the orders are again 2 and 4.

n	mean only		mean and first moments	
	R_{PP}	R_{PS}	R_{PP}	R_{PS}
1	9.3×10^{-4}	1.7×10^{-3}	7.2×10^{-8}	4.1×10^{-7}
2	2.3×10^{-4}	4.2×10^{-4}	4.5×10^{-9}	2.5×10^{-8}
8	1.5×10^{-5}	2.7×10^{-5}	1.7×10^{-11}	9.9×10^{-11}
order	2	2	4	4

583

584 second order and the first-moment voxel fourth order in both the reflected

585 P and the converted S wave: the breakdown of the tiling identity at $O(k_{\parallel}h)$
586 and the evanescent coupling of the gradient channels do not cost an order.
587 At $\mathbf{k}_{\parallel} = 0$ the construction reproduces Eq. (32) to all printed digits.

588 *Incident S.* Nothing in the construction is specific to P. For a normally in-
589 cident S wave the layer is the one-dimensional problem of Eq. (32) with
590 $M_P \rightarrow \mu$, $k_P \rightarrow k_S$ and $\Delta M \rightarrow \Delta\mu$; the collocation and mean-only schemes
591 converge at order 2.00 and the first-moment voxel at 4.00, in reflection and
592 transmission, and the full 9×9 construction with an incident SH wave re-
593 produces that one-dimensional result to all printed digits. At oblique inci-
594 dence, against the exact layer (`ContinuumLimit_IncidentS.wl`), the orders
are those of Table 5. The orders, and with $k = k_S$ the closed-form constants

Table 5: Observed order of convergence ($n = 2 \rightarrow 8$, $\omega = 300 \text{ rad s}^{-1}$) for incident SV and SH waves, with the mean-only and first-moment voxels. The SV angles lie below the critical angle $\arcsin(\beta/\alpha) = 36.9^\circ$.

	mean only	mean and first moments
SV at 20° : R_{SS} / R_{SP}	2.00 / 2.00	4.00 / 3.99
SV at 30° : R_{SS} / R_{SP}	1.98 / 2.00	3.97 / 4.00
SH at 20° and 40° : R_{SH}	2.00	4.00

595

596 of Eqs. (30) and (33), are those of the incident P wave.

597 The spectral construction is implemented twice. The second implementa-
598 tion (`scripts/crosscheck_first_moment_voxel.py`) shares only the defi-
599 nition of the scheme: it takes the depth dependence of the transformed kernel
600 from an independent residue construction, obtains the local weight as a nu-
601 merical large- k_z limit of the inverted Christoffel matrix, evaluates every dou-

602 ble integral by quadrature and separates P from S by sampling the upgoing
 603 field at two depths. The raw reflected displacements of the two agree to $9 \times$
 604 10^{-16} in all 36 cases compared: incident P, SV and SH, normal incidence and
 605 20° , $n = 1, 2, 4$, with and without the first moments. The one-dimensional
 606 scheme of any degree is also implemented twice. The second implementation
 607 (`scripts/crosscheck_second_moment_voxel.py`) evaluates every integral
 608 by Gauss quadrature, split at the receiver within a voxel, and the exact layer
 609 from its own interface system, in double precision. Its errors agree with the
 610 40-digit ones within its round-off floor (3×10^{-14} in reflection, 1.6×10^{-13} in
 611 transmission); at $\omega = 1200 \text{ rad s}^{-1}$ it measures orders 4.02 / 3.99 and 6.02 /
 612 6.00; for one plane as the layer thins its differences from the exact layer fall
 613 as $D^{3.00}$, $D^{5.00}$ and $D^{7.00}$ for $p = 0, 1$ and 2 ; and it rederives c_p for $p \leq 5$.
 614 Figure 4(a,b) shows the convergence, with the orders fitted to it.

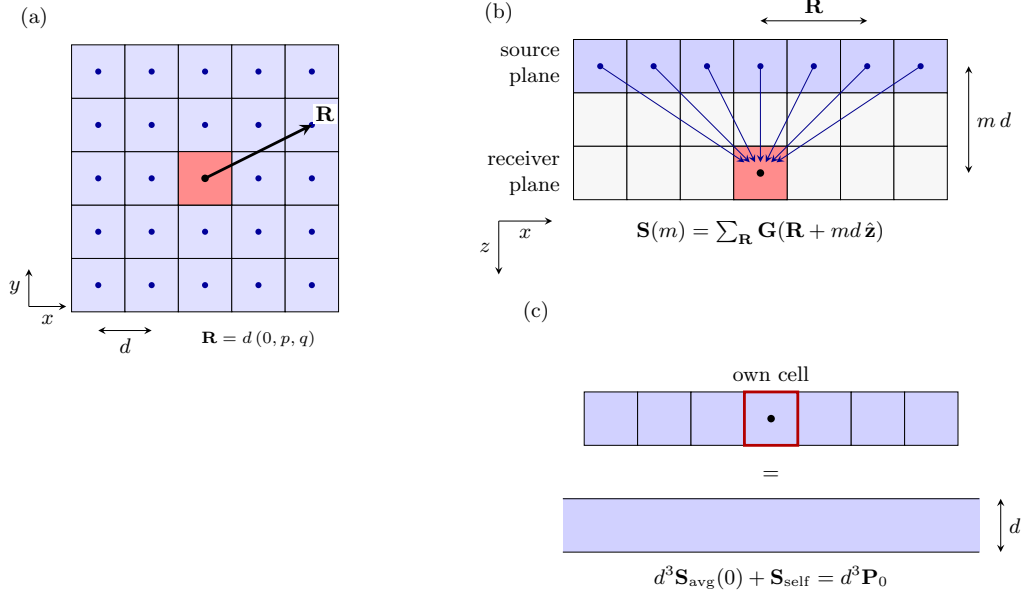


Figure 1: The lattice and its sums. (a) Plan view of one plane of voxels: their centres form a square lattice of pitch d , and a lattice vector $\mathbf{R} = d(0, p, q)$ joins the centre of one voxel (red) to another. (b) Cross-section (z down), between planes: the voxel of a receiver plane (red) is coupled to every voxel of a source plane m planes above it, one for each lattice vector $\mathbf{R}$; the sum over $\mathbf{R}$ is the specular lattice sum of Eq. (6). (c) Within a plane: the sum over every cell of the plane, the receiver's own cell (outlined) included, is the field of a uniform plate of thickness d , the tiling identity of Eq. (13).

Alt text: Three schematic panels. On the left, a plan view of a five-by-five grid of square voxels with a dot at each centre; the centre voxel is highlighted, an arrow labelled $\mathbf{R}$ runs from its centre to another voxel centre, and the spacing between neighbouring centres is marked d . At the top right, a cross-section through three rows of voxels: arrows from the centre of every voxel in the top row converge on a highlighted voxel two rows below, with the plane separation $m d$ marked. Below it, a single row of voxels with one outlined as the receiver's own cell is shown equal to a continuous shaded band of the same thickness d , a uniform plate.

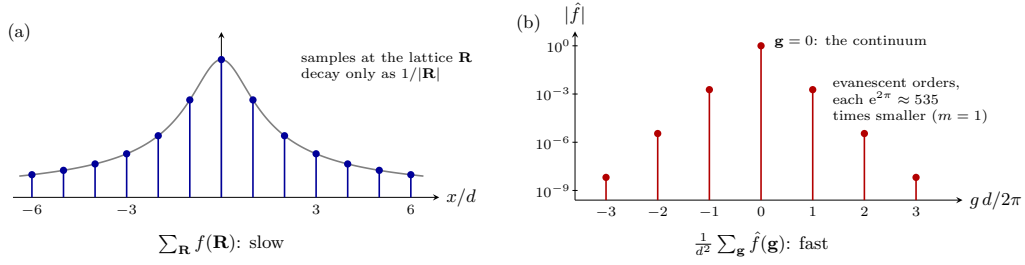


Figure 2: The Poisson summation formula (9) applied to a plane of voxels, along one lateral axis. (a) Real space: the field of each voxel of a source plane at a receiver one plane away, sampled at the lattice points; the samples decay only as $1/|\mathbf{R}|$, so their sum converges slowly. (b) The same sum after Poisson summation, on a logarithmic scale: samples of the lateral Fourier transform at the reciprocal-lattice orders $\mathbf{g} = 2\pi p/d$. The order $\mathbf{g} = 0$ is the continuum's plane wave; every other order is evanescent and, at $m = 1$, smaller than its neighbour by $e^{2\pi} \approx 535$. Magnitudes are schematic: the static decay $e^{-|\mathbf{g}|md}$ is shown.

Alt text: Two panels. On the left, a smooth bell-shaped curve that decays slowly on both sides is sampled by vertical stems at integer positions from minus six to six, the stems remaining clearly visible far from the centre. On the right, on a logarithmic vertical axis from one down to ten to the minus nine, a stem of height one at zero is labelled as the continuum, and stems at plus and minus one, two and three fall by about three powers of ten per step and are labelled as evanescent orders.

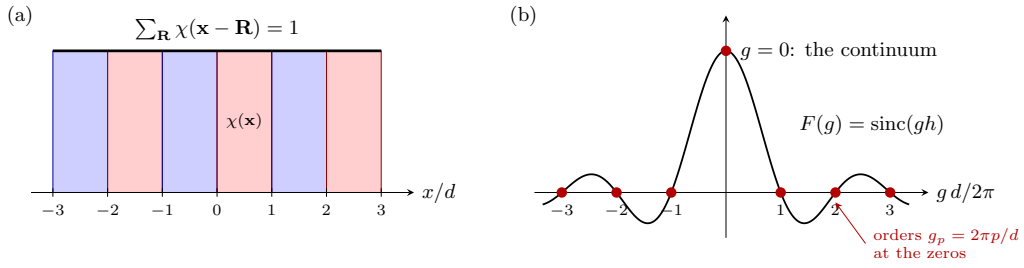


Figure 3: Why cubes make the coupling exact. (a) Cubes tile the plane: the indicator functions χ of the translated cells sum to exactly one everywhere, a partition of unity. (b) The same fact in Fourier space: the cube's lateral form factor $F(g) = \text{sinc}(gh)$ (shown along one axis) vanishes at every reciprocal-lattice order $g_p = 2\pi p/d$ with $p \neq 0$, so of all the orders in the lattice sum (10) only $g = 0$, the continuum's plane wave, survives the source-cell average.

Alt text: Two panels. On the left, adjacent box functions of unit height and unit width, alternately shaded, lie side by side along an axis marked in voxel widths, and a thick horizontal line at height one marks their sum. On the right, the sinc curve, equal to one at the origin and oscillating with decaying amplitude, is plotted against wavenumber in units of $2\pi/d$; dots mark the reciprocal-lattice orders, one at height one at the origin and all the others on the axis at the zeros of the curve.

615 9. A heterogeneous layer

616 A heterogeneous medium is not a layer of identical cells. The simplest
617 heterogeneous case with an exact answer is a stratified model built from layers
618 of constant properties, with real interfaces between them: the form of many
619 layered crustal velocity models. The last test takes such a model. A layer
620 $D = 4$ m thick is divided into eight model layers of 0.5 m, and in each one $\Delta\lambda$,
621 $\Delta\mu$ and $\Delta\rho$ are drawn uniformly within $\pm 15\%$ of the background (the values
622 are listed in `ContinuumLimit_Heterogeneous.w1`). The result is a random
623 stratification with physical contrasts. The voxels refine the model's layering,
624 with m planes of voxels in each model layer, so no voxel straddles a model
625 interface: the natural discretisation when the model's interfaces are known.
626 At $\omega = 1500$ rad s⁻¹ the layer is $k_P D = 1.2$ and $k_S D = 2.0$ thick, so the field
627 reverberates between the model interfaces. The exact answer is the product
628 of the model layers' propagators. Its same-type reflection coefficients agree
629 with those of the reflection–transmission recursion of Kennett [40], computed
630 independently, to 10^{-14} . The schemes are those of §§7–8, with the contrast
631 now changing from plane to plane; nothing else changes. The errors are in
632 Table 6 and Fig. 4(c).

633 *The interfaces cost nothing.* Every order is the one measured for the uniform
634 layer. At a model interface the strain of the true field jumps. A voxel basis
635 that is discontinuous from voxel to voxel represents that jump exactly when
636 the interface lies on a voxel boundary, so the interface adds no error of its
637 own. The only requirement is that voxel boundaries coincide with the model's
638 interfaces. A voxel that straddled an interface would carry a contrast that is
639 not uniform within it, a case not examined here.

Table 6: Relative error against the exact stratified layer, for m planes of voxels per model layer (§9). At normal incidence the order is measured from $m = 4$ to 16, where it is also 2.00 for collocation and 4.00 in transmission; at 20° it is measured from $m = 2$ to 4.

		$m = 1$	$m = 2$	$m = 4$	order
P, normal, R	mean only	1.9×10^{-3}	4.7×10^{-4}	1.2×10^{-4}	2.00
	first moment	8.5×10^{-7}	5.3×10^{-8}	3.3×10^{-9}	4.00
S, normal, R	mean only	5.3×10^{-3}	1.3×10^{-3}	3.3×10^{-4}	2.00
	first moment	6.2×10^{-6}	3.9×10^{-7}	2.4×10^{-8}	4.00
P, 20° , R_{PP}	mean only	1.4×10^{-3}	3.6×10^{-4}	9.0×10^{-5}	2.00
	first moments	3.3×10^{-7}	2.1×10^{-8}	1.3×10^{-9}	4.00
P, 20° , R_{PS}	mean only	2.8×10^{-3}	6.9×10^{-4}	1.7×10^{-4}	2.00
	first moments	1.4×10^{-6}	8.8×10^{-8}	5.5×10^{-9}	4.00

640 *What fourth order buys.* For a reflection error of 10^{-6} at normal incidence
641 (P), the mean-only voxel needs 347 planes of voxels, about 43 per model
642 layer. The first-moment voxel needs 8, one per model layer: the model's own
643 layering, which already gives 8.5×10^{-7} . Counting unknowns (9 per voxel
644 against 36), that is 3123 against 288 per vertical column of voxels, a factor
645 of eleven. In three dimensions, with the lateral grid refined as the vertical
646 one is, the number of voxels falls by $(347/8)^3 \approx 8 \times 10^4$ and the number of
647 unknowns by about 2×10^4 . The same model has been solved independently
648 in Python, with a contrast for each plane: the raw reflected displacements
649 agree with the Mathematica scheme to 1.1×10^{-15} at normal incidence and
650 at 20° , with and without the first moments. Lateral heterogeneity needs the
651 full three-dimensional solver; it is the first extension discussed below.

652 *A medium that varies within a cell.* The stratified model is exact on its own
 653 grid: every cell is homogeneous. A smoothly varying medium is not, and
 654 sampling it as cells of constant properties is an approximation of its own.
 655 Let the contrast of cell j be its Legendre projection to degree r , $\Delta(z) =$
 656 $\Delta_0 \sum_{c \leq r} f_c^{(j)} P_c(s/h)$, with the voxel's field of degree p as in §8. The product
 657 of contrast and field is again a Legendre sum, $P_c P_b = \sum_e A_{cbe} P_e$, with $A_{cbe} =$
 658 $(2e+1) \begin{pmatrix} c & b & e \\ 0 & 0 & 0 \end{pmatrix}^2$ the square of a Wigner $3j$ symbol, so the coupling of Eq. (32)
 659 becomes

$$\int_{\mathbf{i}} \int_j \phi_a(z) \mathbf{K}(z-z') \Delta(z') \phi_b(z') dz dz' = \sum_{c \leq r} \sum_{e \leq p+r} f_c^{(j)} A_{cbe} \left[\int_{\mathbf{i}} \int_j \phi_a(z) \mathbf{K}(z-z') \phi_e(z') dz dz' \right] \Delta_0 : \quad (36)$$

660 the same kernel moments, now to degree $p+r$, and nothing else new. Against
 661 the exact graded layer, a high-precision solution of $(M(z)u')' + \omega^2 \rho(z)u = 0$,
 662 the order of convergence is

$$\min(2p+2, 2r+2), \quad (37)$$

663 or $2p+2$ when degree r represents the profile exactly (Table 7; `ContinuumLimit_`
 664 `GradedContrast.wl`). A contrast that is constant in each cell therefore
 665 caps every voxel at second order, however rich its field basis, and for the
 666 smooth profile the voxel of degree 2 with a linear contrast has the error
 667 of the voxel of degree 1 to within 0.06%. At first order in the contrast
 668 this is what one expects: the contrast's projection error is orthogonal to
 669 the polynomials of degree r , so against the smooth product of the inci-
 670 dent and adjoint fields it contributes at $O(h^{2r+2})$. To keep order $2p+2$
 671 the contrast must be represented to the degree of the field. The closed forms
 672 of the construction are tested one by one by an independent implementa-

Table 7: A contrast that varies within the cells: the field of each voxel to Legendre degree p , its contrast to degree r . Relative reflection error at $n = 16$ planes and the order ($n = 4 \rightarrow 8 \rightarrow 16$), $\omega = 300 \text{ rad s}^{-1}$, for the profiles $1 + (2z/D - 1)/2$ (linear) and $1 + \frac{1}{2} \sin(2\pi z/D)$ (smooth) times the contrast of §7. Predicted: Eq. (37), and $2p + 2$ for the linear profile once $r \geq 1$.

p	r	linear profile		smooth profile	
		error	order	error	order
0	0	7.7×10^{-5}	2.00		
1	0	7.6×10^{-5}	2.00	5.5×10^{-5}	1.95
1	1	6.4×10^{-11}	4.00	1.4×10^{-7}	3.95
2	1	2.3×10^{-17}	6.00	1.4×10^{-7}	3.95
2	2			1.6×10^{-10}	5.98

tion (`scripts/crosscheck_graded_contrast.py`): the Legendre moments
of the kernel against $2h i^n j_n(ch)$ and 30-digit quadrature (to 2×10^{-16}), the
linearisation coefficients against the $3j$ symbols (exactly), and the same-cell
double integrals against 30-digit quadrature (1.4×10^{-16}). It then assem-
bles the scheme without linearisation, evaluating the projected contrast at
the quadrature nodes, against its own 30-digit solution of the graded layer;
its errors agree with those of Table 7 to 10^{-3} , and at $\omega = 1200 \text{ rad s}^{-1}$ it
measures the orders of Eq. (37) again.

10. A sphere: what the layer predicts in three dimensions

The layer and the stratified layer are laterally uniform. A sphere is not,
and elastic scattering by a sphere, homogeneous [41, 42] or radially varying
[43], has an exact solution: it is the usual benchmark for volume-integral and

685 discrete-dipole methods. This section solves two spheres with the full three-
 686 dimensional voxel scheme: the sphere is voxelised on an n_{sub}^3 grid, the Foldy-
 687 Lax system is solved with the cell-averaged coupling by an FFT-accelerated
 688 iterative solver, and a P wave is incident. For the graded sphere the voxel is
 689 the mean-only voxel of §8 in three dimensions: each cell carries the mean of its
 690 nine-component state, tested with the same constant, and the L^2 projection
 691 of its contrast, which is its cell mean; every cell that overlaps the sphere is
 692 kept, and the cells are coupled by cell-to-cell double integrals of the Green's
 693 tensor, one block per orbit of the cube group [2]. For the sharp sphere
 694 the voxel is the package's cube T -matrix, which differs from the collocation
 695 closure only through its far-field corrections (§6.4) and has the same order,
 696 each cell keeping the contrast of its centre. The far field is computed from
 697 the solution for the incident plane wave itself, each voxel's displacement and
 698 strain carrying the plane-wave phase at its centre; adding to that phase a
 699 linear variation about the scatterer's centre, as a Taylor expansion of the
 700 incident field would, counts the wave's variation twice and leaves an error of
 701 order $(ka)^2$ that no refinement removes. The error is the largest difference
 702 between the scattered far field and the exact one over nine scattering angles
 703 from 11° to 169° , relative to the largest exact value, at a distance of 5×10^8
 704 radii for the graded sphere: at 5×10^4 radii the near-field terms that the
 705 asymptotic far field omits are 1.4×10^{-4} of the peak, 4% of the discretisation
 706 error at 12 voxels across and most of it at 24. The sharp sphere, whose errors
 707 are a hundred times larger, is observed at 5×10^4 radii.

708 *A sphere without a staircase.* A voxelised sphere with a sharp surface changes
 709 its shape with every grid. A sphere whose contrast falls smoothly to zero at

710 its surface does not: the voxels that cross the surface carry a vanishing con-
 711 trast. The body here has a homogeneous core $r < a/2$ carrying the contrast
 712 of §7, and across the shell $a/2 < r < a$ the contrast falls to zero as the
 713 smoothstep $s(x) = 10x^3 - 15x^4 + 6x^5$, $x = (a - r)/(a - a/2)$, which vanishes
 714 like $(a - r)^3$ at the surface. Its exact scattering follows, order by order, from
 715 the radial equations of Takeuchi and Saito [43], integrated across the shell
 716 from the core's regular solutions and matched to outgoing waves at $r = a$.
 717 The equations are derived symbolically from the equations of motion, and
 718 the solution is checked against a sphere of homogeneous concentric shells
 719 in closed form, by unitarity and reciprocity, and by an independent imple-
 720 mentation that agrees to 2×10^{-11} (`TakeuchiSaito.wl`, `ContinuumLimit_`
 721 `GradedSphere.wl`, `scripts/crosscheck_graded_sphere.py`). Each voxel
 722 carries the mean of the contrast over its cell. Table 8 shows the error falling
 723 at second order from $n_{\text{sub}} = 6$ to 24 at both frequencies (`scripts/pilot_`
 724 `graded_voxel_sphere.py`): the apparent order between successive grids is
 725 2.02 to 2.06 at $k_S a = 0.5$ and 2.00 at $k_S a = 1$, and the least-squares orders
 726 of Fig. 4(d) are 2.045 ± 0.004 and 1.994 ± 0.004 . With 24 voxels across the
 727 error is 1.8×10^{-4} and 8.3×10^{-4} . At $n_{\text{sub}} = 4$ a voxel is as wide as the
 728 graded shell and the order is not yet asymptotic. This is the order the layer
 729 predicts for the mean-only voxel, and the first test of that prediction on a
 730 body with genuine three-dimensional structure.

731 What makes the sequence this clean is that the voxel carries a single error.
 732 The collocation voxel of the sharp sphere applied to this body, each cell at
 733 the profile's value at its centre and only the cells whose centres lie inside the
 734 sphere kept, carries two: its own internal-field error and the error of sampling

735 the profile, both $O(h^2)$. Its error falls from 1.0×10^{-1} at $n_{\text{sub}} = 4$ to 4.7×10^{-3} ,
 736 2.6×10^{-3} , 6.4×10^{-4} and 2.3×10^{-4} at 6, 8, 12 and 16, apparent orders 7.6,
 737 2.0, 3.5 and 3.6 at $k_{Sa} = 0.5$, and at $k_{Sa} = 1$ the apparent orders are 1.7, 2.5
 738 and 2.4 from 6 to 16: on these grids its error is below the mean-only voxel's,
 739 but its apparent order is not that of any single term and cannot be read from
 740 them. The same collocation voxel with the cell mean of the contrast, and
 741 every overlapping cell kept, gives 3.4×10^{-3} , 1.9×10^{-3} and 8.2×10^{-4} at 6,
 742 8 and 12, apparent orders 2.03 and 2.05: the sampling of the profile was the
 743 second term, and with it removed the collocation and mean-only voxels differ
 744 only by their constants, as in Table 3. The cell mean, with every overlapping
 745 cell kept, represents the same body on every grid, and the order the layer
 746 predicts is then the one observed.

747 The first-moment voxel of §8, extended to three dimensions with a con-
 748 trast linear in each cell, converges at fourth order on this same sphere [2]: its
 749 apparent orders are 4.01 and 4.13 where the discretisation error is isolated
 750 at weak contrast, and at the full contrast its error with 16 voxels across is
 751 4.9×10^{-6} and 5.9×10^{-6} , 82 and 317 times below the mean-only voxel's on
 752 the same grid.

753 *A sphere with a sharp surface.* The homogeneous sphere is the familiar test,
 754 and its voxelisation changes with every grid. Table 9 shows its error for grids
 755 of 4, 6 and 8 voxels across: 4.6%, then 19%, then 4.2% at $k_{Sa} = 0.5$. The
 756 error tracks the staircase's volume, which is 5%, 17% and 4% too large on
 757 these grids. Rescaled by the ratio of the sphere's volume to the staircase's,
 758 it falls below 1% at $k_{Sa} = 0.5$ and to at most 3% at $k_{Sa} = 1.0$, but it still
 759 does not converge: what remains is the staircase's shape. A refinement study

Table 8: The smoothly graded sphere, voxelised with the mean-only voxel on grids of n_{sub} voxels across, against its exact solution: relative far-field error, and the apparent order between successive grids. Nine unknowns per cell, every cell overlapping the sphere kept.

n_{sub}	unknowns	$k_{\text{ga}} = 0.5$		$k_{\text{ga}} = 1.0$	
		error	order	error	order
4	576	7.70×10^{-3}		3.24×10^{-2}	
6	1 872	3.04×10^{-3}	2.30	1.32×10^{-2}	2.22
8	3 672	1.68×10^{-3}	2.05	7.52×10^{-3}	1.95
12	11 520	7.30×10^{-4}	2.06	3.34×10^{-3}	2.00
16	24 984	4.06×10^{-4}	2.04	1.88×10^{-3}	2.00
20	47 304	2.58×10^{-4}	2.03	1.20×10^{-3}	2.00
24	79 344	1.79×10^{-4}	2.02	8.34×10^{-4}	2.00

760 of a sharp sphere measures its staircase; the smooth sphere, like the layer,
761 measures the scheme.

Table 9: A homogeneous sphere voxelised on grids of n_{sub} voxels across, against exact elastic scattering: relative far-field error, raw and rescaled by the ratio of the sphere’s volume to the staircase’s. It does not converge with refinement, because the staircase changes with the grid.

n_{sub}	voxels	$k_{\text{sa}} = 0.5$		$k_{\text{sa}} = 1.0$	
		raw	volume-rescaled	raw	volume-rescaled
4	32	4.6×10^{-2}	1.4×10^{-3}	5.0×10^{-2}	5.4×10^{-3}
6	136	1.94×10^{-1}	7.9×10^{-3}	1.68×10^{-1}	3.0×10^{-2}
8	280	4.2×10^{-2}	2.3×10^{-3}	3.5×10^{-2}	8.9×10^{-3}

762 11. Discussion

763 Figure 4 collects the convergence of every test, with the order fitted to
764 each by least squares (`scripts/plot_convergence_orders.py`).

765 *Where the error of a voxel scheme lives.* The results separate the two ap-
766 proximations a voxel scheme makes. The *coupling* between voxels makes
767 none: averaged over the source cell it is the continuum’s cell integral ex-
768 actly, between planes by the form factor’s zeros and within a plane by the
769 tiling identity, and the self term it omits is exactly the one the single site
770 carries. Every error therefore comes from the *single site*, from the internal
771 field it assumes. For a uniform field that error is the midpoint rule’s, known
772 in closed form; for a linear field it is the next term of the same expansion.
773 The converse is the practical warning. A kernel that is not the cell integral
774 (the point Green’s tensor, or a cell average that does not match the single
775 site’s closure) leaves a local term that does not vanish as the voxels shrink
776 (§4): the evanescent lattice orders carry a static contribution of fixed size per

777 voxel volume. A scheme built that way converges to the wrong limit, and
 778 refinement cannot reveal it, because the error is scale-invariant.

779 *What the single site is.* The moment hierarchy of §6 turns the uniform-field
 780 single site from a modelling choice into a derived object with a stated error.
 781 Its static part is exact for a sphere, and for a cube it differs from the sphere’s
 782 only by the cubic split of Eq. (26). Its dynamic part departs from exact
 783 scattering by a sphere only through the real $(ka)^2$ term of Eq. (27); the
 784 radiation damping is exact. That term and the collocation error of Eq. (30)
 785 have the same origin, the variation of the field across the scatterer, and the
 786 same remedy, a richer internal field. In a lattice the remedy is cheap: the
 787 field a voxel must represent is the smooth continuum field, not the field of
 788 an isolated cube, with its concentrations at the edges and corners.

789 *Relation to the coupled-dipole method.* The same partition appears in the
 790 coupled-dipole method as the choice of polarisability. The Clausius–Mossotti
 791 relation, the lattice dispersion relation [9], the polarisability taken from the
 792 exact static solution [10] and the integrated Green tensor [11] all adjust
 793 the single site so that the lattice reproduces the continuum. For space-
 794 filling cubes the adjustment is neither needed nor a free parameter: the tiling
 795 identity fixes the local field exactly, the self term of the collocation closure is
 796 the one that makes it cancel, and what is left is a closed-form error rather than
 797 an empirical one. That the method is a collocation of the volume integral
 798 equation was shown for the electromagnetic case by Lakhtakia [7]; Eq. (29)
 799 is the same statement for the elastic lattice with the self term evaluated over
 800 the finite cube, and it is what makes the error a midpoint-rule error. The

801 filtered coupled dipoles of Piller and Martin [12] and Yurkin et al. [13] band-
 802 limit the source to the grid; the source-cell average does the complementary
 803 thing exactly, its form factor vanishing on the reciprocal lattice, and the
 804 spectral construction of Eq. (35) is the periodic coupled-dipole method of
 805 Chaumet et al. [14] with the Galerkin double average in place of the point
 806 dipole. The second-order convergence reported for the DDA by Yurkin et al.
 807 [1] corresponds to the uniform-field voxel's, and the opposite-signed constants
 808 of Eq. (30) explain why a polarisability correction can improve one observable
 809 only at the expense of another, as the package's own form-factor corrections
 810 do in reflection and transmission. Going beyond second order requires a richer
 811 internal field, which is the point of higher-order bases for volume integral
 812 equations; the first-moment voxel is the lowest such basis, and in its spectral
 813 form the Galerkin double average makes the lattice sum converge without
 814 Ewald splitting.

815 *The tension between the two error sources.* The coupled-dipole method has
 816 two sources of error, the single site and the coupling between sites, and each
 817 has been corrected on its own: the polarisability by the Clausius–Mossotti
 818 relation, the lattice dispersion relation or the exact static limit, and the
 819 coupling by integrating or filtering the Green tensor. The two are not in-
 820 dependent. The point kernel's evanescent orders carry a static local term
 821 whose size per voxel volume does not shrink with the voxels (§4), and the
 822 single site must cancel exactly that term. A polarisability tuned for one
 823 kernel is therefore wrong for another: the lattice dispersion relation corrects
 824 the point-dipole lattice, and changing the kernel to an integrated or a fil-
 825 tered one changes the term it must cancel. Tuning either alone moves the

826 error between the two, and refinement cannot reveal a mismatch, because
 827 the surviving term is scale-invariant. For space-filling cubes the tension is
 828 resolved rather than balanced: the cell-averaged kernel omits exactly the self
 829 term that the closure carries (Eq. (29)), so the coupling error vanishes iden-
 830 tically and every remaining error is the single site's, the variation of the field
 831 across the cell. That error has a second tension of its own. It is a midpoint-
 832 rule error whose constants have opposite signs in reflection and transmission
 833 (Eq. (30)), so any scalar adjustment of the single site that improves one wors-
 834 ens the other; the package's far-field corrections reduce the reflection error
 835 by a factor 4.45 and increase the transmission error by 1.7. No polarisability,
 836 however chosen, can remove it. What removes it is a richer internal field:
 837 the first moment lowers the error by two orders in kd in both observables at
 838 once.

839 *The cost of fourth order.* A first-moment voxel carries four times as many
 840 unknowns as a uniform one (36 against 9 in three dimensions). The closed
 841 forms show what that buys. To reach an error ε in reflection, the uniform
 842 voxel needs $kd = \sqrt{8\varepsilon}$ and the first-moment voxel $kd = (720\varepsilon)^{1/4}$; at $\varepsilon = 10^{-6}$
 843 that is $kd = 2.8 \times 10^{-3}$ against 0.16, voxels 58 times larger in each direction.
 844 In a three-dimensional model the voxel count falls by $58^3 \approx 2 \times 10^5$ and the
 845 number of unknowns by about 5×10^4 (these are the Born-order constants;
 846 at the contrast studied the first-moment voxel's measured errors lie within
 847 5% of them, while the uniform voxel's transmission constant grows with the
 848 contrast, so the comparison is if anything conservative). The stratified model
 849 of §9 measures it directly: 347 planes of uniform-field voxels against 8 first-
 850 moment voxels for the same accuracy. Higher moments continue the trend

851 in one dimension: by Eq. (34) the second-moment voxel reaches $\varepsilon = 10^{-6}$ at
 852 $kd = 0.68$ and the third-moment voxel at $kd = 1.5$, where a voxel is already
 853 a quarter of a wavelength. A three-dimensional voxel of degree p has not
 854 been built; with the Legendre polynomials of total degree at most p it would
 855 carry 10 ($p = 2$) or 20 ($p = 3$) weights per state component, against 4 for
 856 the first-moment voxel. The gain is largest precisely where it matters: when
 857 the model varies on scales that are coarse compared with the accuracy the
 858 wavefield must have.

859 *Limitations and outlook.* The results concern laterally uniform media, a ho-
 860 mogeneous layer and a randomly stratified one, for which the discrete system
 861 is exactly a chain of planes; a laterally heterogeneous medium needs the first-
 862 moment voxel in a full three-dimensional solver, where the gradient channels
 863 couple through the evanescent lattice orders (§8) and the tiling identity holds
 864 only to $O(k_{\parallel}h)$. The smoothly graded sphere of §10 shows the mean-only
 865 voxel converging at second order on a laterally heterogeneous body against
 866 an exact solution, and the companion paper [2] shows the first-moment voxel
 867 converging at fourth order there. A contrast that varies within a cell, which
 868 §9 shows is needed beyond second order in a smoothly varying medium, has
 869 a further consequence in three dimensions: its linear part is odd about the
 870 cell's centre, so it couples the single site's uniform-field and first-moment
 871 blocks, which parity keeps apart for a homogeneous cube (§6.3). A graded
 872 voxel therefore needs at least the first-moment block; that voxel is built,
 873 and tested on the graded sphere, in the companion paper. The numerical
 874 evidence is at moderate, physically typical contrasts (up to 15% in density
 875 and moduli) and at $k_{\parallel}h \leq 0.3$, the range in which the single-site T -matrix is

876 validated; the closed forms are first order in the contrast, and the departures
 877 from them at the full contrast are of second order. What is established is the
 878 *order* of convergence and its leading constant, not the behaviour at strong
 879 contrast, where the single site's own closure may limit the accuracy before
 880 the lattice coupling does. The single site's dynamic error is in closed form
 881 for a sphere only; for a cube, for which no exact solution exists, it is known
 882 through the layer (§§7–8). Two extensions follow directly. The first is the
 883 three-dimensional solver itself. The second is a stratified *background*, a lay-
 884 ered reference medium, as distinct from the stratified contrast of §9. There
 885 the lattice coupling then carries the reverberation of the layering, which is
 886 not translation invariant in depth, and whether a tiling identity holds for it
 887 is the natural next question.

888 **12. Conclusions**

889 A lattice of cubic voxels reproduces a continuous elastic medium in a pre-
 890 cise sense. Because cubes tile the plane, the source-cell-averaged coupling be-
 891 tween voxels is the continuum's cell integral exactly, and at normal incidence
 892 the lattice kernel is available in closed form with no lattice sum. The single
 893 site is derived rather than assumed. The moments of the Green's tensor over
 894 the cube, taken as distributions, give a closed hierarchy whose first-gradient
 895 truncation is the uniform-strain closure. That closure is exact for a sphere in
 896 the static limit, and it departs from exact scattering by a sphere only by a
 897 real $(ka)^2$ term, with the radiation damping exact. For a cube it differs from
 898 the sphere only by a cubic anisotropy of the shear response. The collocation
 899 closure carries the same self term as the lattice kernel, which cancels, so the

900 discrete system is a collocation of the continuum equation. It converges at
 901 second order, with a leading error that is a midpoint-rule error of the internal
 902 field, known in closed form for every discretisation, and opposite in sign in
 903 reflection and transmission. A voxel that also carries the first moment of
 904 its internal field converges at fourth order, at normal incidence (where the
 905 three-dimensional scheme reduces exactly to one dimension) and at oblique
 906 incidence, with a leading error, $-(kd)^4/720$, that no longer depends on the
 907 direction of observation. With Legendre moments to degree p it converges at
 908 order $2p + 2$ with the closed-form constant of Eq. (34), and a single plane of
 909 such voxels reproduces the thin-layer series term by term through $O(D^{2p+2})$.
 910 Where the medium varies within a cell, a contrast sampled as constant caps
 911 every voxel at second order; with the contrast projected to degree r the order
 912 is $\min(2p + 2, 2r + 2)$, and the coupling needs nothing beyond the same kernel
 913 moments to degree $p + r$. The orders survive the interfaces of a randomly
 914 stratified model, provided voxel boundaries coincide with its interfaces, and
 915 there the first-moment voxel reaches an accuracy of 10^{-6} with one voxel
 916 per model layer, where the uniform-field voxel needs about forty. A sphere
 917 whose contrast falls smoothly to zero at its surface converges at second order
 918 against its exact solution, as the layer predicts, to 0.018% at $k_S a = 0.5$ with
 919 twenty-four voxels across; a sharp sphere's error is set by its staircase, which
 920 changes with the grid. The results, and the evidence for each, are collected
 921 in Table 10.

Table 10: The results of the paper, and the evidence for each.

Result	Evidence
Continuous layer, exact and to $O(D^k)$	reference to 2.5×10^{-8} ; series order k
Thin-layer series, Eq. (7)	each coefficient symbolically, to D^4
Point-kernel local term survives $d \rightarrow 0$	$11 \times$ the continuum term at $m = 1$
Averaged coupling between planes = continuum $\times$ sinc	2.8×10^{-16} , 81 entries
Tiling identity, Eq. (13)	21/21 components to 4×10^{-9}
Exact kernel, Eq. (14)	replaces a 9.4×10^{-4} static error
Moments as distributions, Eq. (21)	sum rule (22) exact; M, Q equal an independent archive, 81/81
Channels (23)–(25)	a production code by another route, 2×10^{-15}
Cube vs sphere, Eq. (26)	symbolic; and numerically at five ν , 4×10^{-16}
Ball closure = exact Mie, static	2×10^{-10} ; cube's bulk and shear average 10^{-8}
Ball closure vs Mie, dynamic, Eq. (27)	symbolic; numerical Mie 5×10^{-4} ; $(ka)^3$ terms exact
Self term cancels, Eq. (29)	4×10^{-16}
Package T = closure statically	2×10^{-5} at $k_S h = 0.01$
Second-order convergence	order 2.000, three frequencies
Closed-form error, Eq. (30)	$n = 1$ –16, to 10^{-4} of the error
First-moment voxel, Eq. (32)	order 4.00, three frequencies
First-moment voxel, oblique, Eq. (35)	order 4, P and S, 20° and 40°

922 **Acknowledgments**

923 This work began thirty years ago in the author’s doctoral research at
924 the Research School of Earth Sciences of the Australian National University,
925 Canberra [17].

926 **Declaration of competing interest**

927 The author declares no known competing financial interests or personal
928 relationships that could have appeared to influence the work reported in this
929 paper.

930 **Declaration of generative AI and AI-assisted technologies in the** 931 **manuscript preparation process**

932 During the preparation of this work, the author used Claude (Anthropic)
933 in order to check the existing Mathematica notebooks, Python and Fortran
934 code, and to build tests to ensure their correctness. After using this tool, the
935 author reviewed and edited the content as needed and takes full responsibility
936 for the content of the published article.

937 **Data availability**

938 The Mathematica notebooks and Python scripts that derive and check ev-
939 ery result, and the package they test, are archived at [https://doi.org/10.](https://doi.org/10.5281/zenodo.23003241)
940 5281/zenodo.23003241 and developed at [https://github.com/tmnestor/](https://github.com/tmnestor/MultipleScatteringCalculations)
941 MultipleScatteringCalculations.

942 **References**

- 943 [1] M. A. Yurkin, V. P. Maltsev, A. G. Hoekstra, Convergence of the dis-
944 crete dipole approximation. I. Theoretical analysis, *J. Opt. Soc. Am. A*
945 23 (2006) 2578–2591. doi:10.1364/JOSAA.23.002578.
- 946 [2] T. M. Nestor, Graded voxels in three dimensions: A Galerkin first-
947 moment cube with a contrast linear in the cell, fourth order in its dis-
948 cretisation, 2026. Preprint, archived with this paper at [https://doi.](https://doi.org/10.5281/zenodo.23003241)
949 [org/10.5281/zenodo.23003241](https://doi.org/10.5281/zenodo.23003241).
- 950 [3] L. L. Foldy, The multiple scattering of waves. I. General theory of
951 isotropic scattering by randomly distributed scatterers, *Phys. Rev.* 67
952 (1945) 107–119. doi:10.1103/PhysRev.67.107.
- 953 [4] M. Lax, Multiple scattering of waves, *Rev. Mod. Phys.* 23 (1951) 287–
954 310. doi:10.1103/RevModPhys.23.287.
- 955 [5] E. M. Purcell, C. R. Pennypacker, Scattering and absorption of light
956 by nonspherical dielectric grains, *Astrophys. J.* 186 (1973) 705–714.
957 doi:10.1086/152538.
- 958 [6] B. T. Draine, P. J. Flatau, Discrete-dipole approximation for scattering
959 calculations, *J. Opt. Soc. Am. A* 11 (1994) 1491–1499. doi:10.1364/
960 JOSAA.11.001491.
- 961 [7] A. Lakhtakia, Strong and weak forms of the method of moments and
962 the coupled dipole method for scattering of time-harmonic electromag-
963 netic fields, *Int. J. Mod. Phys. C* 3 (1992) 583–603. doi:10.1142/
964 S0129183192000385.

- 965 [8] M. A. Yurkin, A. G. Hoekstra, The discrete dipole approximation: an
966 overview and recent developments, *J. Quant. Spectrosc. Radiat. Transfer*
967 106 (2007) 558–589. doi:10.1016/j.jqsrt.2007.01.034.
- 968 [9] B. T. Draine, J. Goodman, Beyond Clausius–Mossotti: wave propaga-
969 tion on a polarizable point lattice and the discrete dipole approximation,
970 *Astrophys. J.* 405 (1993) 685–697. doi:10.1086/172396.
- 971 [10] A. Rahmani, P. C. Chaumet, G. W. Bryant, Coupled dipole method
972 with an exact long-wavelength limit and improved accuracy at finite fre-
973 quencies, *Opt. Lett.* 27 (2002) 2118–2120. doi:10.1364/OL.27.002118.
- 974 [11] P. C. Chaumet, A. Sentenac, A. Rahmani, Coupled dipole method
975 for scatterers with large permittivity, *Phys. Rev. E* 70 (2004) 036606.
976 doi:10.1103/PhysRevE.70.036606.
- 977 [12] N. B. Piller, O. J. F. Martin, Increasing the performance of the coupled-
978 dipole approximation: a spectral approach, *IEEE Trans. Antennas*
979 *Propag.* 46 (1998) 1126–1137. doi:10.1109/8.718567.
- 980 [13] M. A. Yurkin, M. Min, A. G. Hoekstra, Application of the discrete dipole
981 approximation to very large refractive indices: filtered coupled dipoles
982 revived, *Phys. Rev. E* 82 (2010) 036703. doi:10.1103/PhysRevE.82.
983 036703.
- 984 [14] P. C. Chaumet, A. Rahmani, G. W. Bryant, Generalization of the
985 coupled dipole method to periodic structures, *Phys. Rev. B* 67 (2003)
986 165404. doi:10.1103/PhysRevB.67.165404.

- 987 [15] M. A. Yurkin, M. Kahnert, Light scattering by a cube: accuracy limits of
988 the discrete dipole approximation and the T-matrix method, *J. Quant.*
989 *Spectrosc. Radiat. Transfer* 123 (2013) 176–183. doi:10.1016/j.jqsrt.
990 2012.10.001.
- 991 [16] J. E. Gubernatis, E. Domany, J. A. Krumhansl, Formal aspects of the
992 theory of the scattering of ultrasound by flaws in elastic materials, *J.*
993 *Appl. Phys.* 48 (1977) 2804–2811. doi:10.1063/1.324141.
- 994 [17] T. M. Nestor, *Seismic Wave Propagation in Complex Media: Sum-*
995 *ming the Multiple Scattering Series*, Ph.D. thesis, Research School of
996 *Earth Sciences*, Australian National University, Canberra, ACT, Aus-
997 *tralia*, 1996. Open access: <https://hdl.handle.net/1885/144359>.
- 998 [18] S. K. Kanaun, V. M. Levin, Scattering of elastic waves on a heteroge-
999 neous inclusion of arbitrary shape: an efficient numerical method for 3D-
1000 problems, *Wave Motion* 50 (2013) 687–707. doi:10.1016/j.wavemoti.
1001 2013.01.002.
- 1002 [19] J. Yang, A. Abubakar, P. M. van den Berg, T. M. Habashy, F. Reitich,
1003 A CG-FFT approach to the solution of a stress-velocity formulation of
1004 three-dimensional elastic scattering problems, *J. Comput. Phys.* 227
1005 (2008) 10018–10039. doi:10.1016/j.jcp.2008.07.027.
- 1006 [20] T. Touhei, A fast volume integral equation method for elastic wave
1007 propagation in a half space, *Int. J. Solids Struct.* 48 (2011) 3194–3208.
1008 doi:10.1016/j.ijsolstr.2011.07.013.

- 1009 [21] M. Jakobsen, T-matrix approach to seismic forward modelling in the
1010 acoustic approximation, *Stud. Geophys. Geod.* 56 (2012) 1–20. doi:10.
1011 1007/s11200-010-9081-2.
- 1012 [22] M. Jakobsen, I. Pšenčík, E. Iversen, B. Ursin, Transition operator ap-
1013 proach to seismic full-waveform inversion in arbitrary anisotropic elastic
1014 media, *Commun. Comput. Phys.* 28 (2020) 297–327. doi:10.4208/cicp.
1015 oa-2018-0197.
- 1016 [23] J. D. Eshelby, The determination of the elastic field of an ellipsoidal
1017 inclusion, and related problems, *Proc. R. Soc. Lond. A* 241 (1957) 376–
1018 396. doi:10.1098/rspa.1957.0133.
- 1019 [24] K. Aki, P. G. Richards, *Quantitative Seismology*, 2nd ed., University
1020 Science Books, Sausalito, CA, 2002.
- 1021 [25] E. M. Stein, G. Weiss, *Introduction to Fourier Analysis on Euclidean*
1022 *Spaces*, Princeton University Press, Princeton, NJ, 1971.
- 1023 [26] C. M. Linton, Lattice sums for the Helmholtz equation, *SIAM Rev.* 52
1024 (2010) 630–674. doi:10.1137/09075130X.
- 1025 [27] J. D. Eshelby, The elastic field outside an ellipsoidal inclusion, *Proc. R.*
1026 *Soc. Lond. A* 252 (1959) 561–569. doi:10.1098/rspa.1959.0173.
- 1027 [28] T. Mura, *Micromechanics of Defects in Solids*, 2nd ed., Martinus Nijhoff,
1028 Dordrecht, 1987. doi:10.1007/978-94-009-3489-4.
- 1029 [29] Y. Mikata, S. Nemat-Nasser, Elastic field due to a dynamically

- transforming spherical inclusion, *J. Appl. Mech.* 57 (1990) 845–849.
doi:10.1115/1.2897650.
- [30] T. M. Michelitsch, H. Gao, V. M. Levin, Dynamic Eshelby tensor and potentials for ellipsoidal inclusions, *Proc. R. Soc. Lond. A* 459 (2003) 863–890. doi:10.1098/rspa.2002.1054.
- [31] Y. P. Chiu, On the stress field due to initial strains in a cuboid surrounded by an infinite elastic space, *J. Appl. Mech.* 44 (1977) 587–590. doi:10.1115/1.3424140.
- [32] J. K. Lee, W. C. Johnson, Calculation of the elastic strain field of a cuboidal precipitate in an anisotropic matrix, *Phys. Status Solidi A* 46 (1978) 267–272. doi:10.1002/pssa.2210460134.
- [33] G. J. Rodin, Eshelby’s inclusion problem for polygons and polyhedra, *J. Mech. Phys. Solids* 44 (1996) 1977–1995. doi:10.1016/S0022-5096(96)00066-X.
- [34] H. Nozaki, M. Taya, Elastic fields in a polyhedral inclusion with uniform eigenstrains and related problems, *J. Appl. Mech.* 68 (2001) 441–452. doi:10.1115/1.1362670.
- [35] A. K. Mal, L. Knopoff, Elastic wave velocities in two-component systems, *J. Inst. Math. Appl.* 3 (1967) 376–387. doi:10.1093/imamat/3.4.376.
- [36] S. K. Kanaun, V. M. Levin, *Self-Consistent Methods for Composites*, Vol. 2: Wave Propagation in Heterogeneous Materials, Solid Me-

- 1052 chanics and Its Applications, Springer, Dordrecht, 2008. doi:10.1007/
1053 978-1-4020-6968-0.
- 1054 [37] R.-S. Wu, K. Aki, Scattering characteristics of elastic waves by an elastic
1055 heterogeneity, *Geophysics* 50 (1985) 582–595. doi:10.1190/1.1441934.
- 1056 [38] P. C. Waterman, Matrix theory of elastic wave scattering, *J. Acoust.*
1057 *Soc. Am.* 60 (1976) 567–580. doi:10.1121/1.381130.
- 1058 [39] V. D. Kupradze, Dynamical Problems in Elasticity, volume 3 of *Progress*
1059 *in Solid Mechanics*, North-Holland, Amsterdam, 1963. Edited by I. N.
1060 Sneddon and R. Hill.
- 1061 [40] B. L. N. Kennett, *Seismic Wave Propagation in Stratified Media*, Cam-
1062 bridge University Press, Cambridge, 1983.
- 1063 [41] C. F. Ying, R. Truell, Scattering of a plane longitudinal wave by a
1064 spherical obstacle in an isotropically elastic solid, *J. Appl. Phys.* 27
1065 (1956) 1086–1097. doi:10.1063/1.1722545.
- 1066 [42] Y.-H. Pao, C.-C. Mow, *Diffraction of Elastic Waves and Dynamic Stress*
1067 *Concentrations*, Crane Russak, New York, 1973.
- 1068 [43] H. Takeuchi, M. Saito, Seismic surface waves, in: B. A. Bolt (Ed.),
1069 *Methods in Computational Physics*, volume 11, Academic Press, New
1070 York, 1972, pp. 217–295.

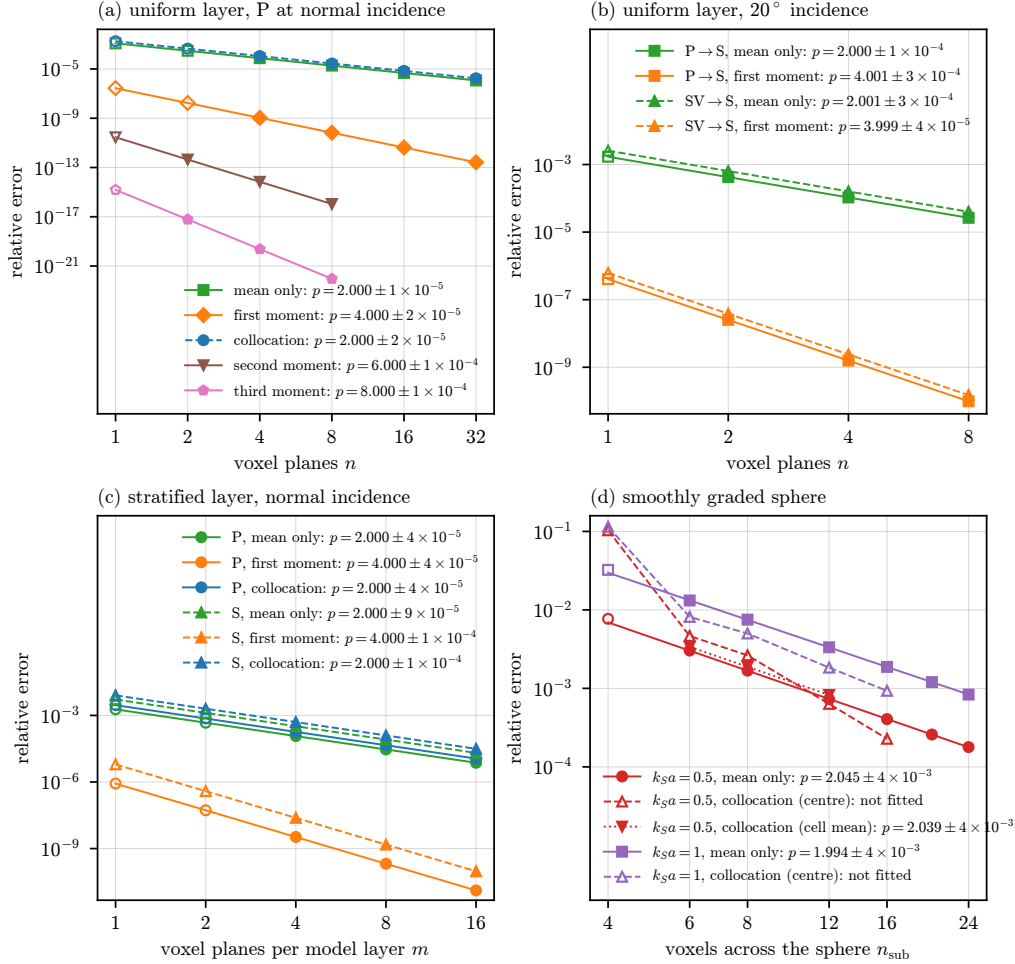


Figure 4: Convergence of every test, with least-squares regression lines. Each line fits $\log e = c - p \log n$ to the filled points; open points are pre-asymptotic and are not fitted. The legend gives the order p and its standard error. Colour marks the scheme: collocation (§6), the mean-only voxel and the voxels carrying Legendre moments to degree $p = 1, 2$ and 3 (Eqs. (32) and (35)). (a) The uniform layer, P reflection at normal incidence ($D = 2 \text{ m}$, $\omega = 300 \text{ rad s}^{-1}$, the contrast of §7), fitted from $n = 4$; the collocation line is dashed and lies on the mean-only line, from which its constant differs only by the factor $3/2$ of Table 3; the second- and third-moment voxels, computed in 40-digit arithmetic, fitted from $n = 2$. (b) The uniform layer at 20° , fitted from $n = 2$: the S wave converted from incident P (solid) and reflected from incident SV (dashed). (c) The stratified layer of §9, reflection at normal incidence for P (solid) and S (dashed), against the number m of voxel planes per model layer, fitted from $m = 4$. (d) The smoothly graded sphere of §10 against its exact solution, for n_{sub} voxels across: the mean-only voxel, fitted from $n_{\text{sub}} = 6$ (at 4 a voxel is as wide as the graded shell); the collocation voxel with the contrast sampled at the cell centre, joined by dashed lines and not fitted, since its two $O(h^2)$ errors of opposite sign leave no single order to fit; and, at $k_{\text{sa}} = 0.5$, the same collocation voxel with the